

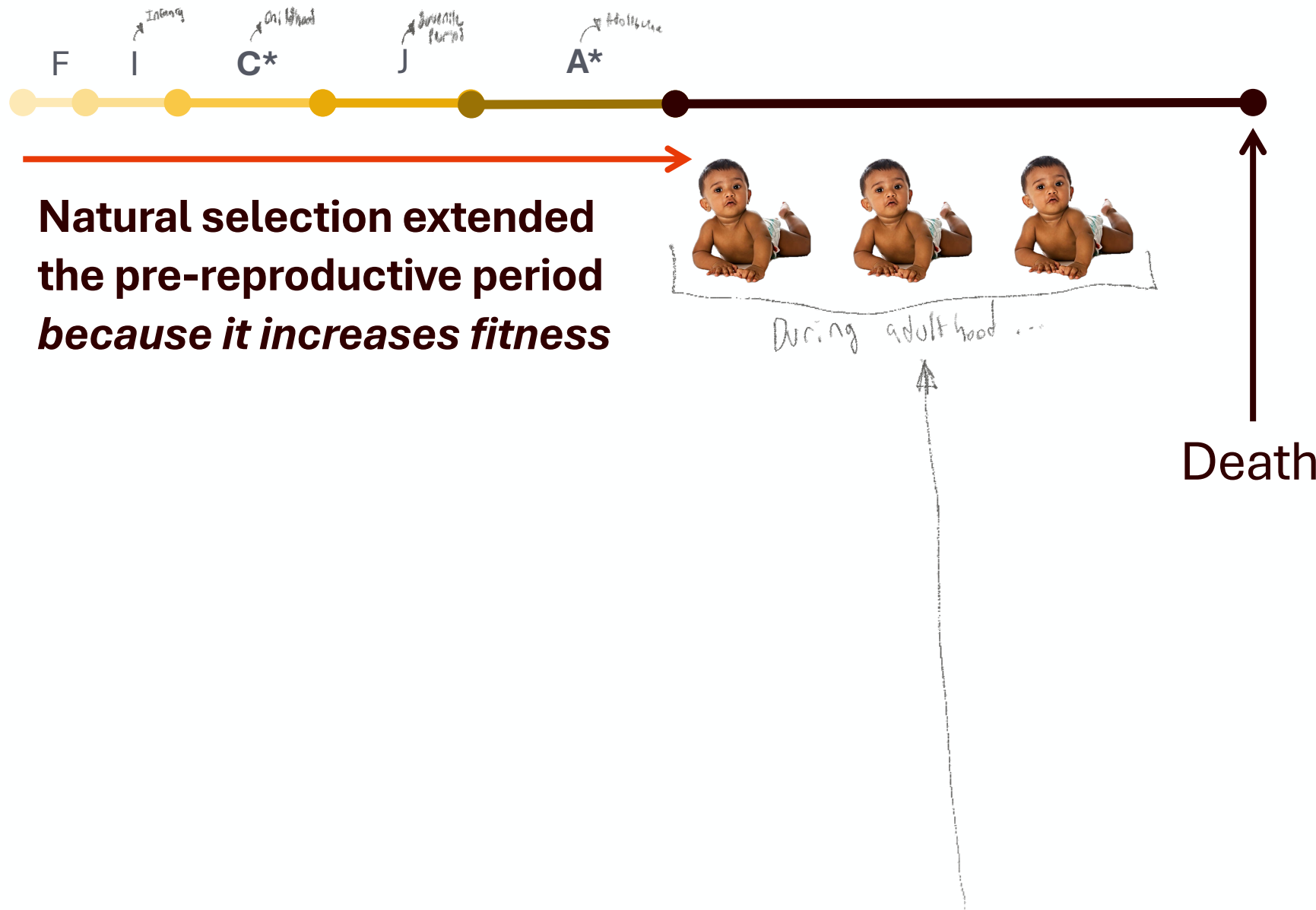


Ageing, senescence, and mortality

Dr. Laura van Holstein

lavanho@emory.edu

ANT215



The point of life is to convert energy to offspring

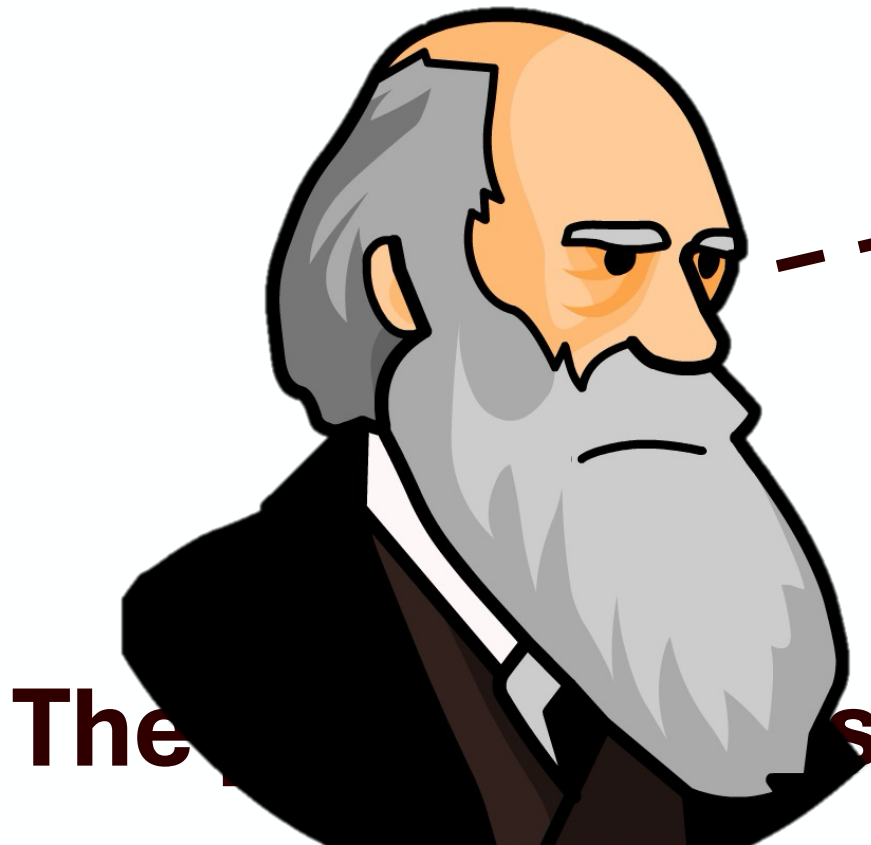


Natural selection extended
the pre-reproductive period
because it increases fitness

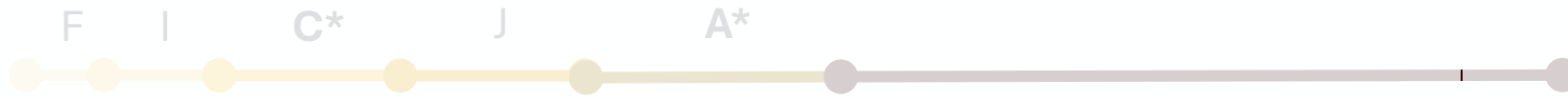


Death

Why do organisms die?
Is death adaptive?



The **goal** is to convert energy to offspring



Different levels where selection can occur

Species

Populations

Individuals

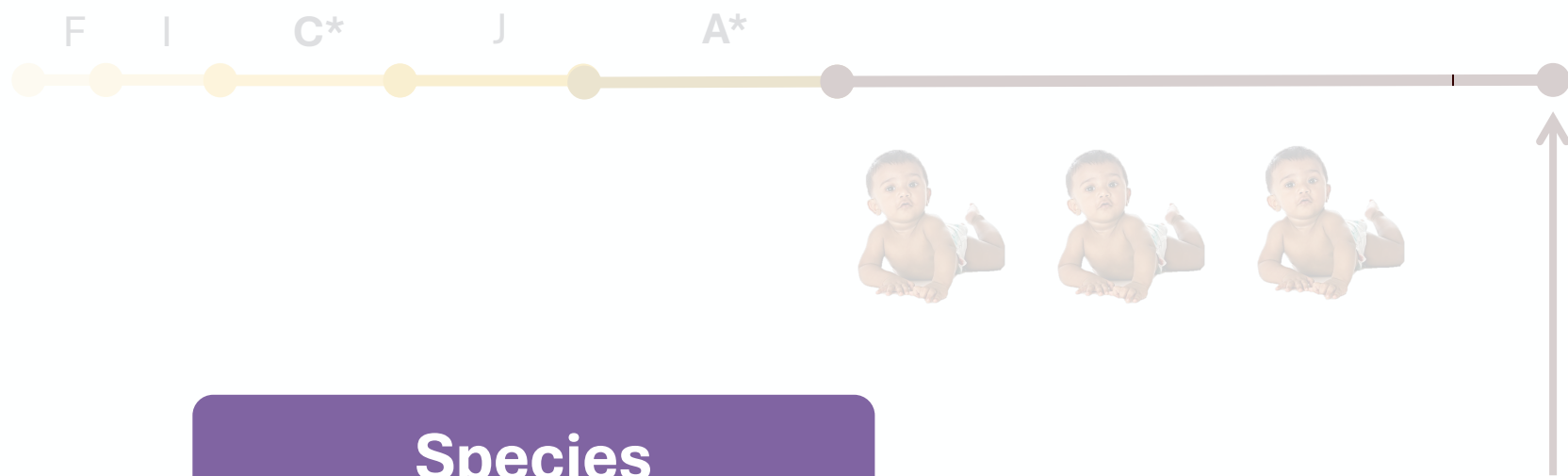
Genes

Death

Why do organisms die?
Is death adaptive?

An individual contains the genes that are passed to the next generation

The organism's job is to convert energy to offspring



Death

Why do organisms die?

Why do organisms live?

Species

Populations

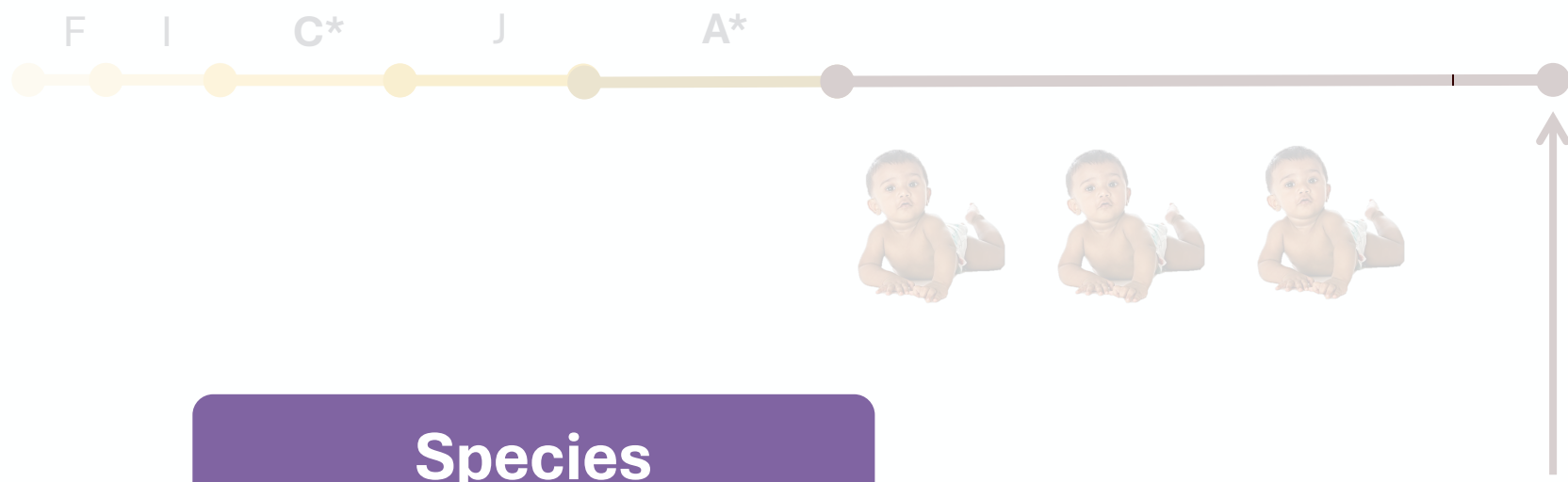
Individuals

Genes



But... continuing to have babies is better than not doing so!

The costs of living



Species

Populations

Individuals

Genes

Death

Why do organisms die?

Why do organisms live?



But... continuing to have babies is better than not doing so!

The costs of living



Weisman (1899):

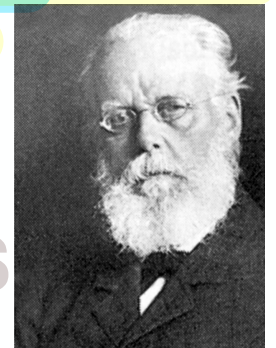
Although ageing is bad for individuals, it **benefits groups** by limiting population density and therefore allowing more space and resources for **younger, fitter, more energetic individuals**

There are huge costs and this is generally viewed to be wrong

But...

- Individual-level selection usually supersedes group selection! *→ Selfish mutant problem*
- In the wild, a “death programme” would only concern a small number of individuals who would have survived
- In most species, age-specific mortality rates increase steadily while in the programmed death scenario a ‘wall of death’ would be expected at a particular age

Because we are not thinking about extrinsic mortality but instead death within the organism



Species

Populations

Individuals

Genes

The ... s to convert energy to offs



Species

Populations

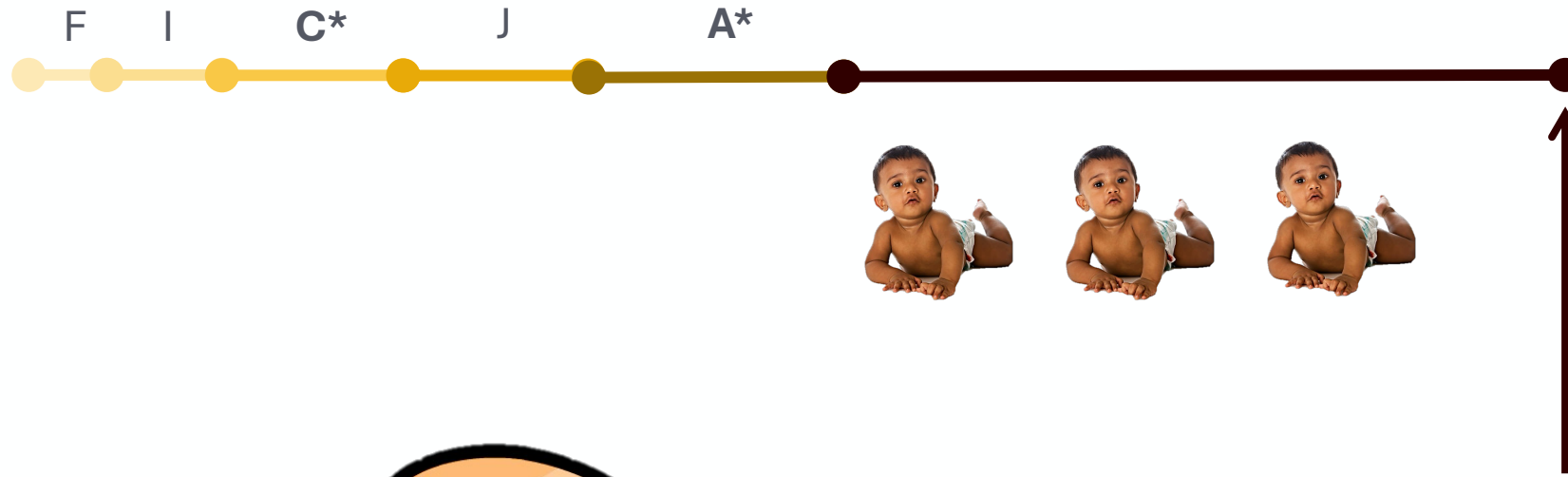
Individuals

Genes

Death

Why do organisms die?
Is death adaptive?

The **task** is to convert energy to offspring



Death

Why do organisms die?
Death is not adaptive.

The  needs to convert energy to offspring



Death

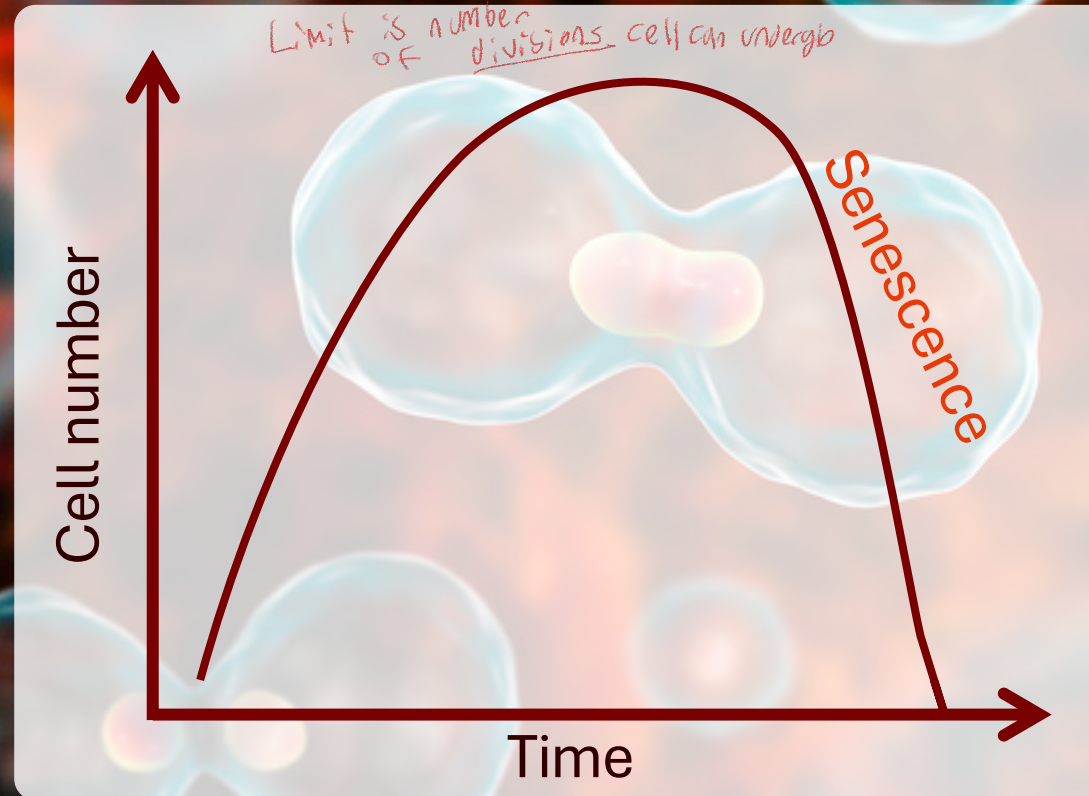
How do organisms die?



The goal is to convert energy to offspring

Hayflick limit

Normal human cells can divide 40-60 times



an individual cell can divide this many times before senescence



Senescence

Accumulation of senescent cells in tissue impairs somatic function

Cells don't function the way they should anymore - you stop being able to maintain homeostasis

Mortality

Hayflick limit:

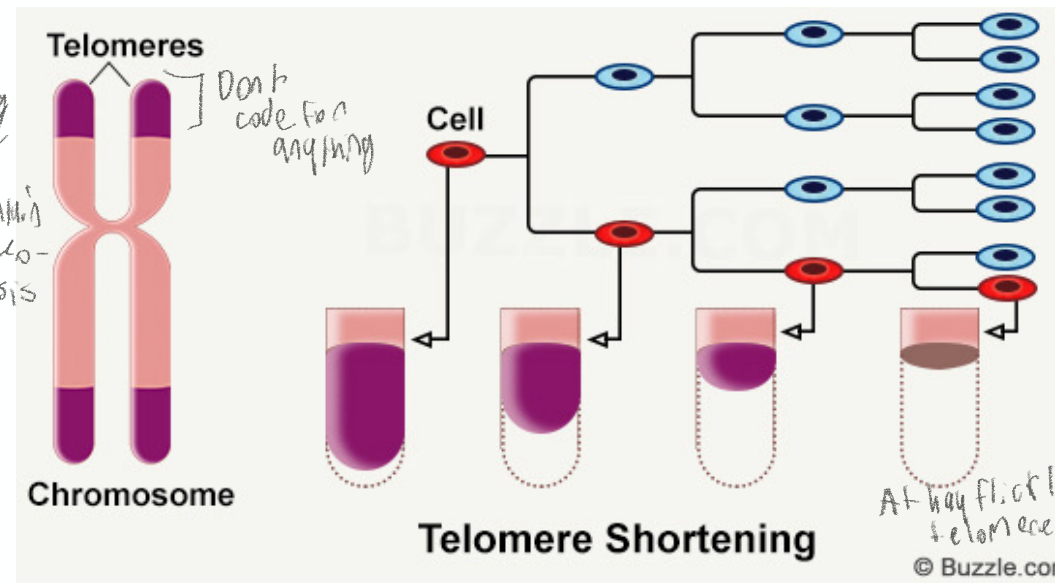
the number of times a normal cell population can divide before entering the replicative senescence phase

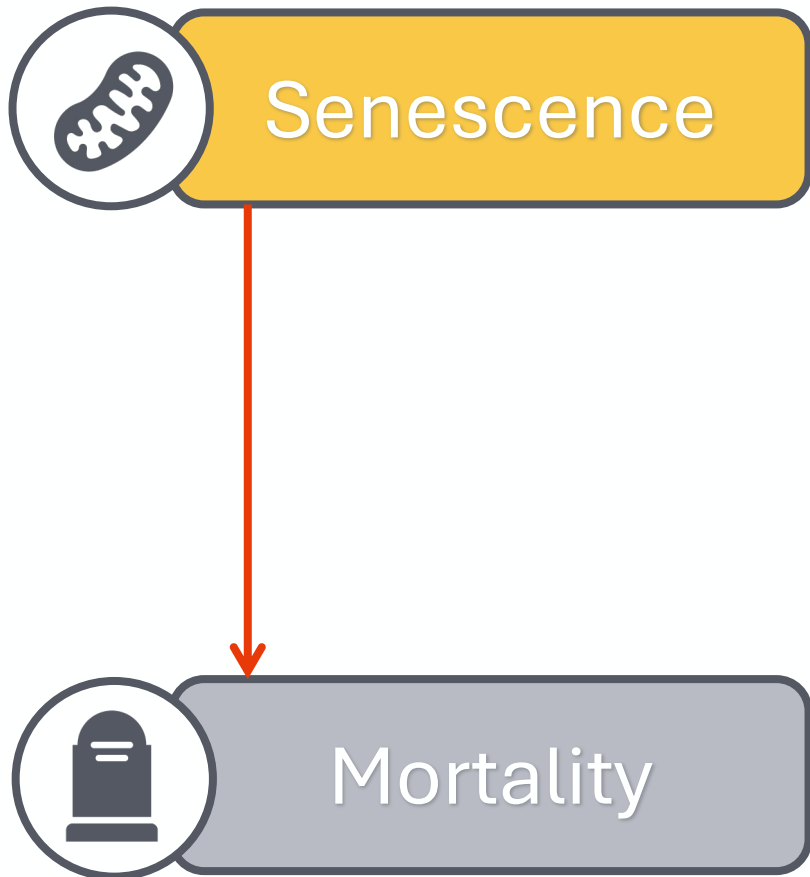
More and more cells become inactive / have trouble operating

Replicative senescence:

the inability of cells to divide. Comes with

- expressing certain proteins
- releasing inflammatory molecules
- loosening tightly packed regions of DNA known as heterochromatin





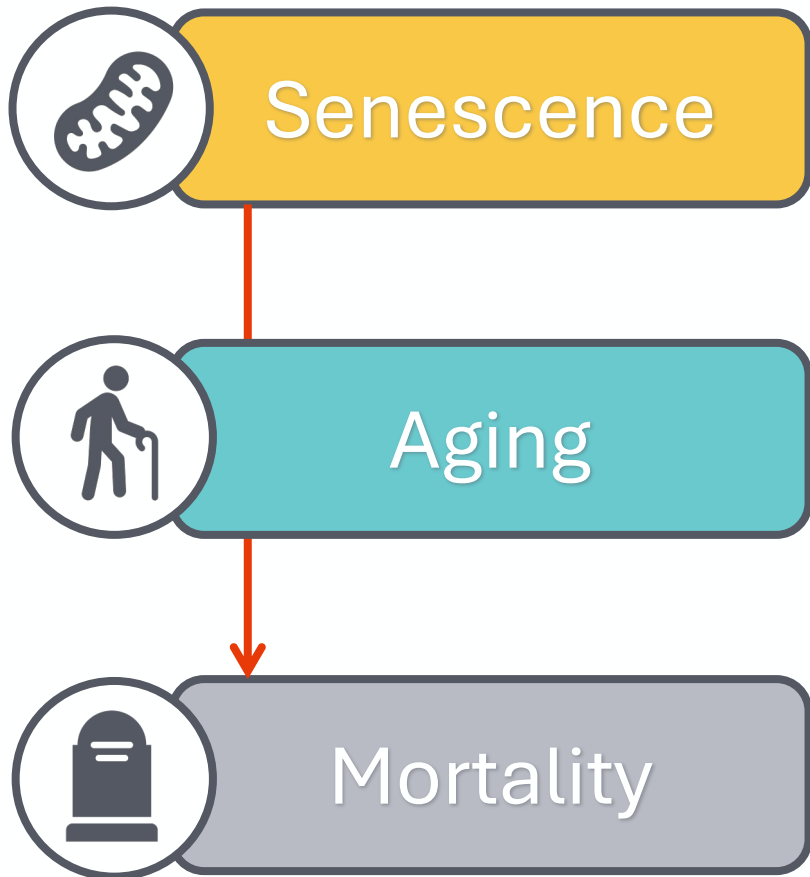
Crews and Ice (2012):

“Senescence is **progressive degeneration** following a period of development and attainment of maximum reproductive potential that **leads to an increased probability of mortality**”

At the organismal level, which is what we care about

It includes:

- **Intrinsic biological processes that lead to declines in cellular function** → previous definition, which is less useful now
- An increased probability of death
- Reduction of reproductive potential over the lifespan



Crews and Ice (2012):

“Senescence is **progressive degeneration** following a period of development and attainment of maximum reproductive potential that **leads to an increased probability of mortality**”

It includes:

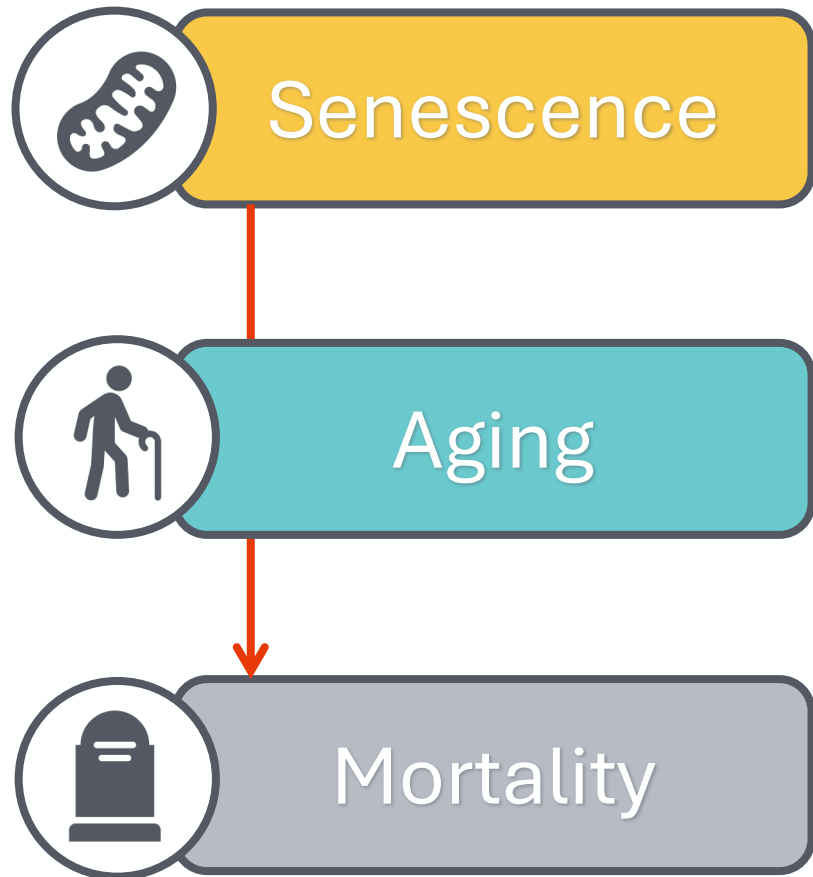
- **Intrinsic biological processes that lead to declines in cellular function**
- **An increased probability of death**
- **Reduction of reproductive potential over the lifespan**

Downstream

functions of

senescence/physical

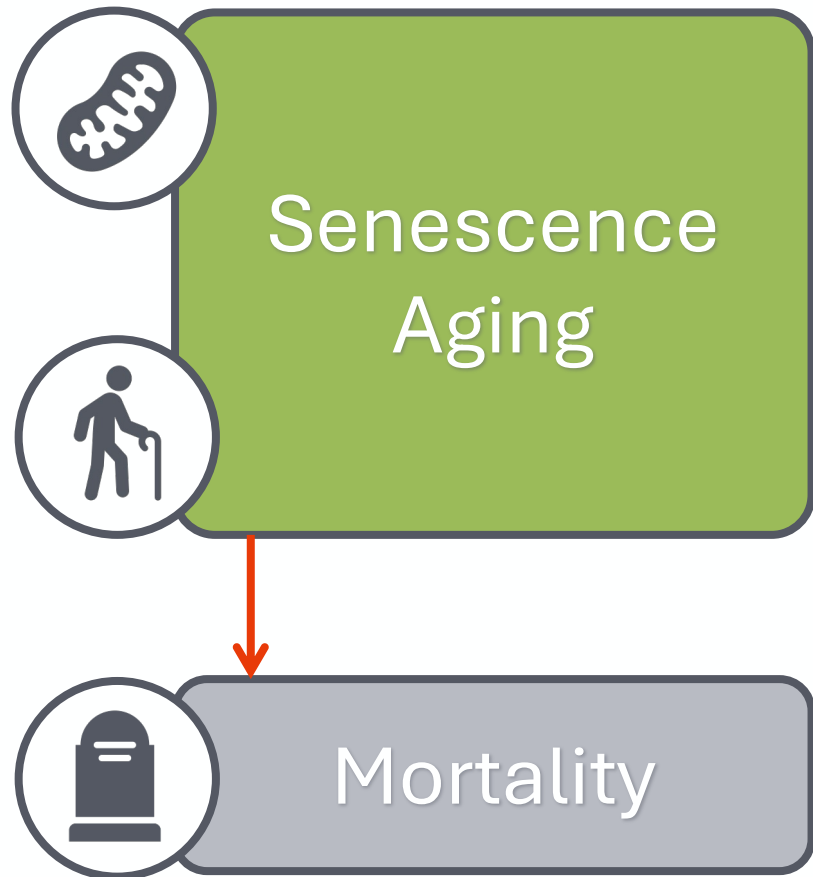
manifestation refers to aging



Crews and Ice (2012):

Ageing: the outward **phenotypic signs** of senescence:

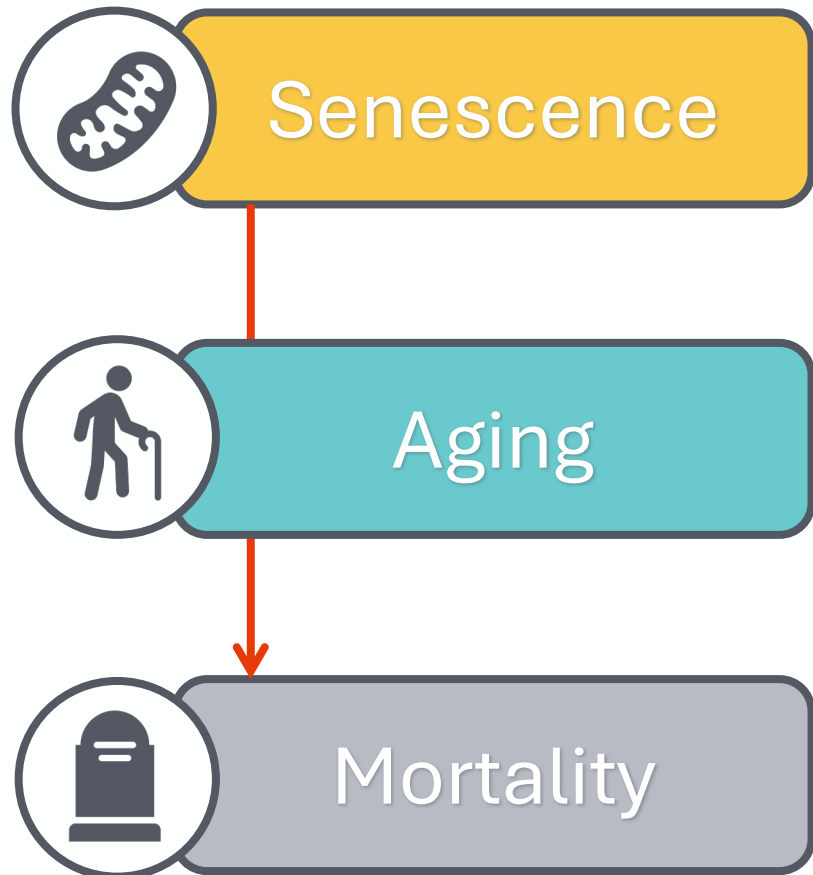
- Increased susceptibility to infectious and chronic diseases
- Loss of resistance to external and internal stressors
- Inabilities to maintain and repair somatic systems
- Progressive increase in frailty



Maklakov and Chapman (2019):

“**Aging, or senescence**, is a physiological deterioration of an organism with advancing age, which reduces reproductive performance and increases the probability of death...”

But not in heritable!

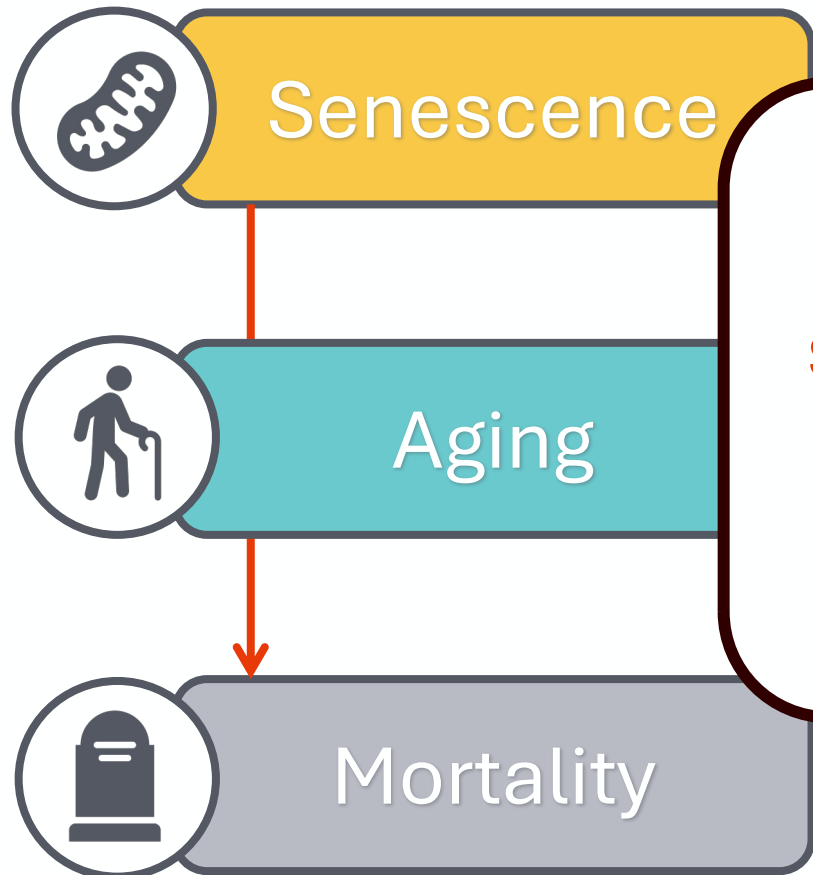


Some of the processes that underlie senescence and ageing:

1. Telomere attrition: telomeres shortening
2. Genomic instability: Excessive DNA damage or insufficient DNA repair *→ Protein and DNA impaired, either wrong or no proteins*
3. Epigenetic alterations: Alterations in the methylation of DNA or acetylation and methylation of histones, as well as of other chromatin-associated proteins
4. Loss of proteostasis: Failure to refold or degrade unfolded proteins *→ Implicated in Alzheimer's?*
5. Deregulated nutrient sensing
6. Mitochondrial dysfunction
7. Stem cell exhaustion
8. Altered intercellular communication: changes at the level of intercellular communication, be it endocrine, neuroendocrine or neuronal

Don't need to know what underlie these just know these processes

Some of the processes that underlie senescence and ageing:

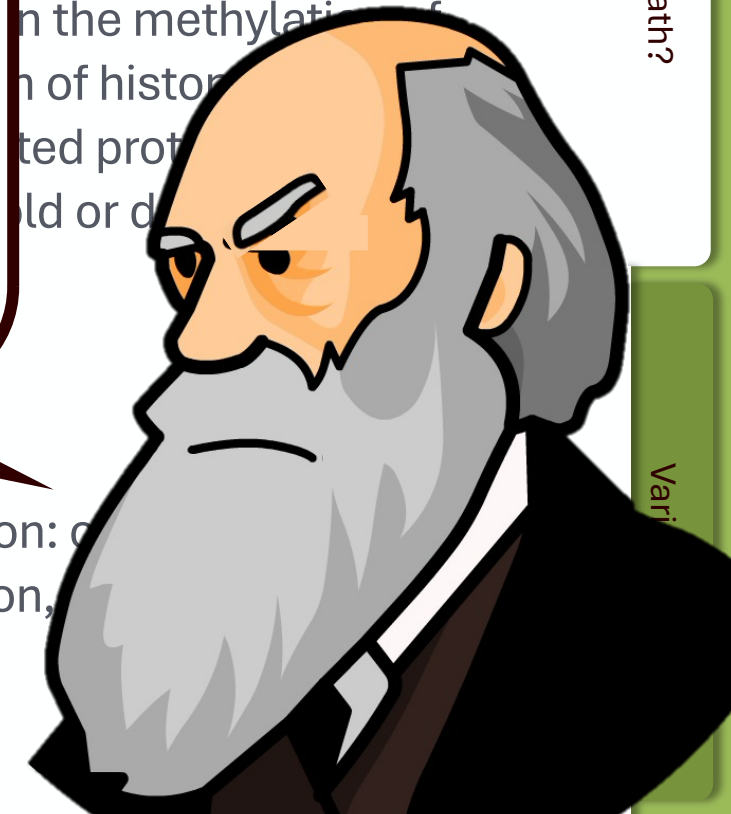


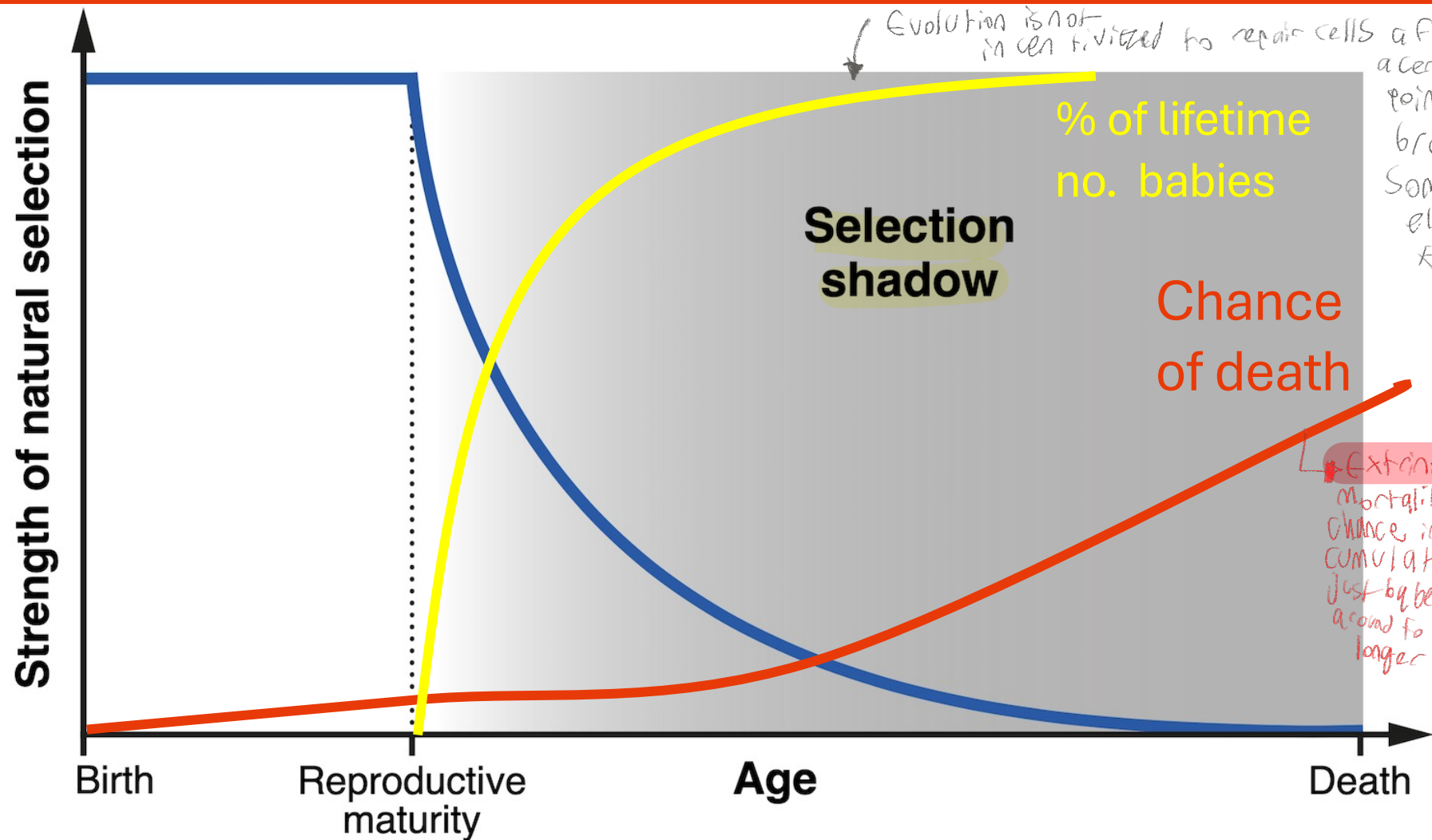
Why has natural selection not selected against these processes?!

- 6. Mitochondrial dysfunction
- 7. Stem cell exhaustion
- 8. Altered intercellular communication: decreased level of intercellular communication, neuroendocrine or neuronal

...ening
...NA damage or

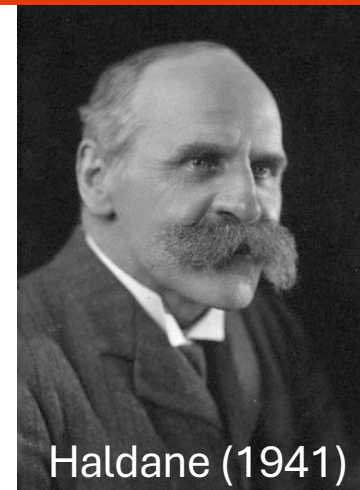
...n the methylation of
...n of histone
...ted proteins
...ld or d



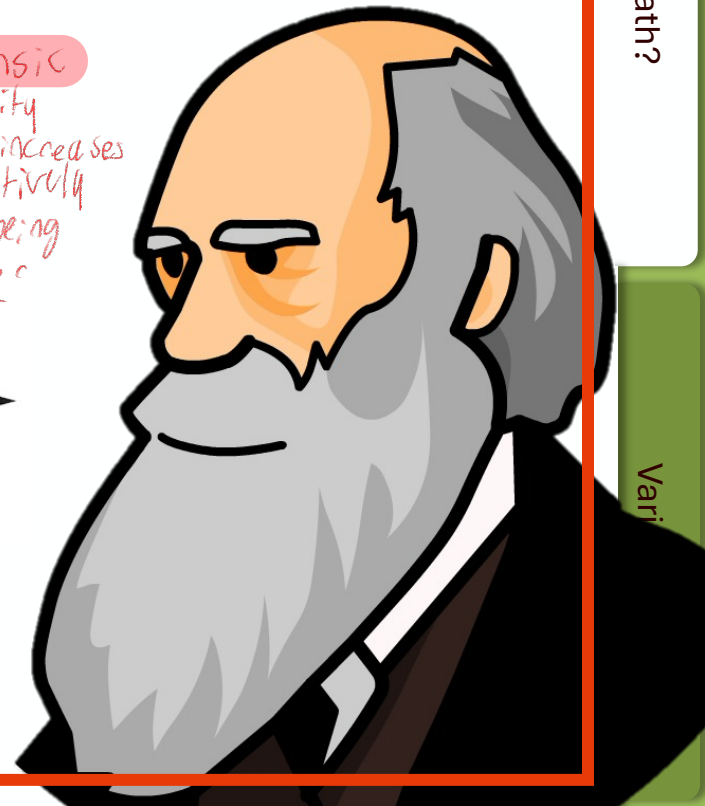


Evolution is not incentivized to repair cells after a certain point bro
Something else might kill you

% of lifetime no. babies



Extrinsic mortality chance increases cumulatively just by being around for a longer

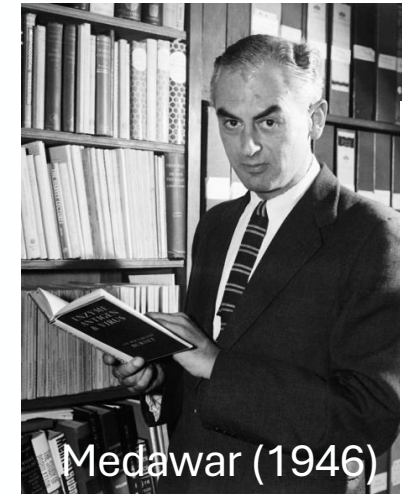
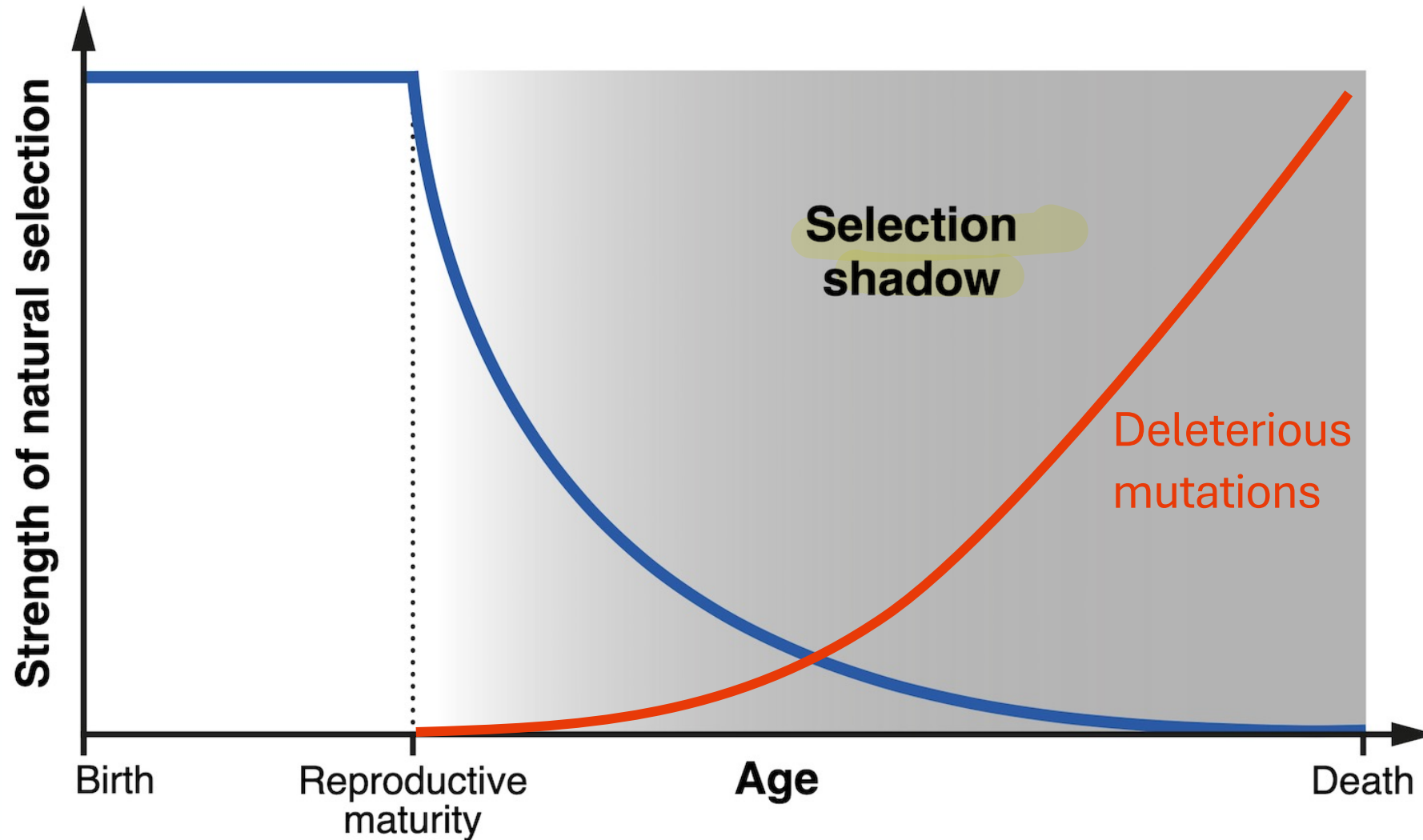


Intro

How death?

Vari

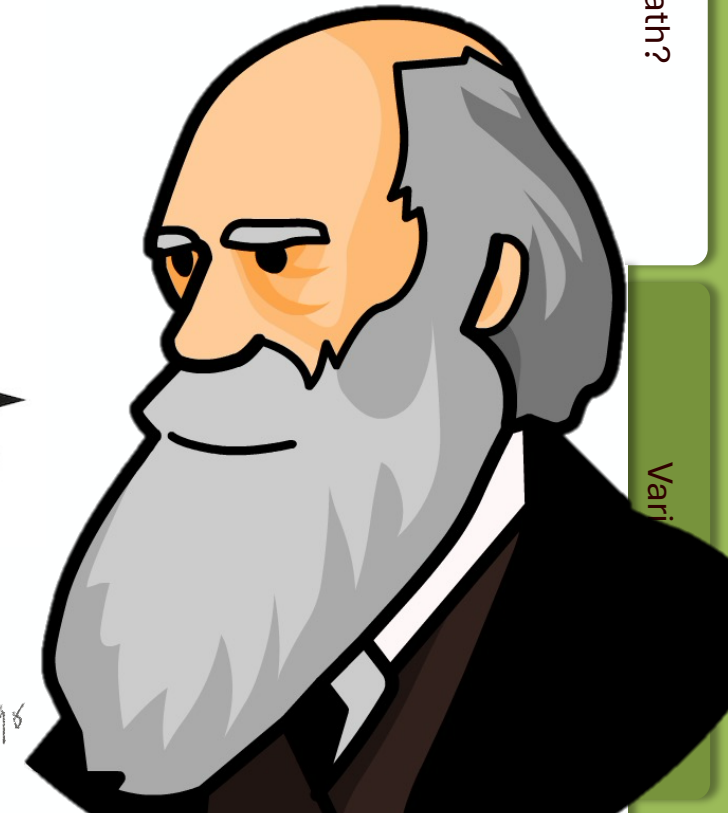
Key

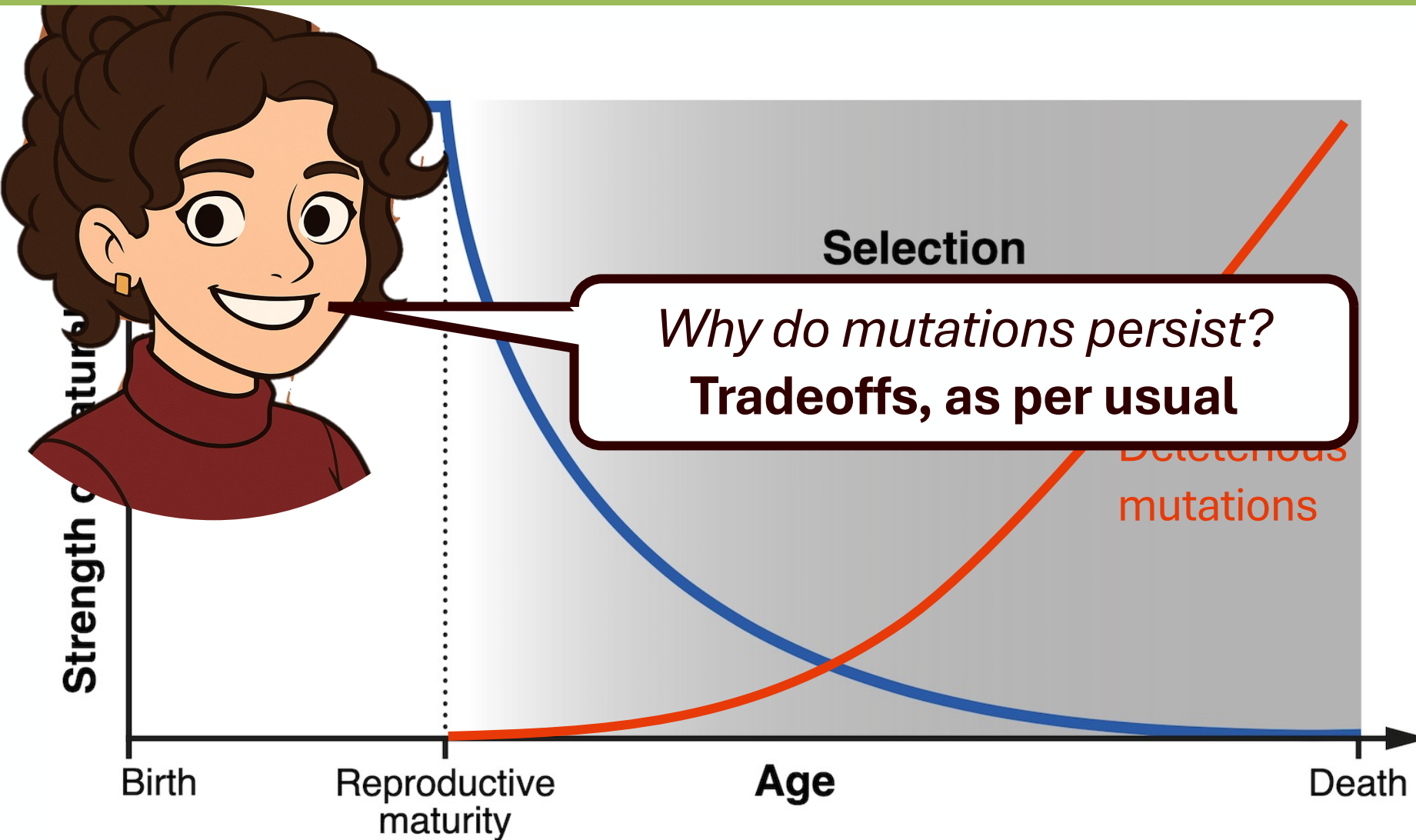


How does this lead to senescence?

because strength of selection decreases, no mechanisms to counter mutations evolves

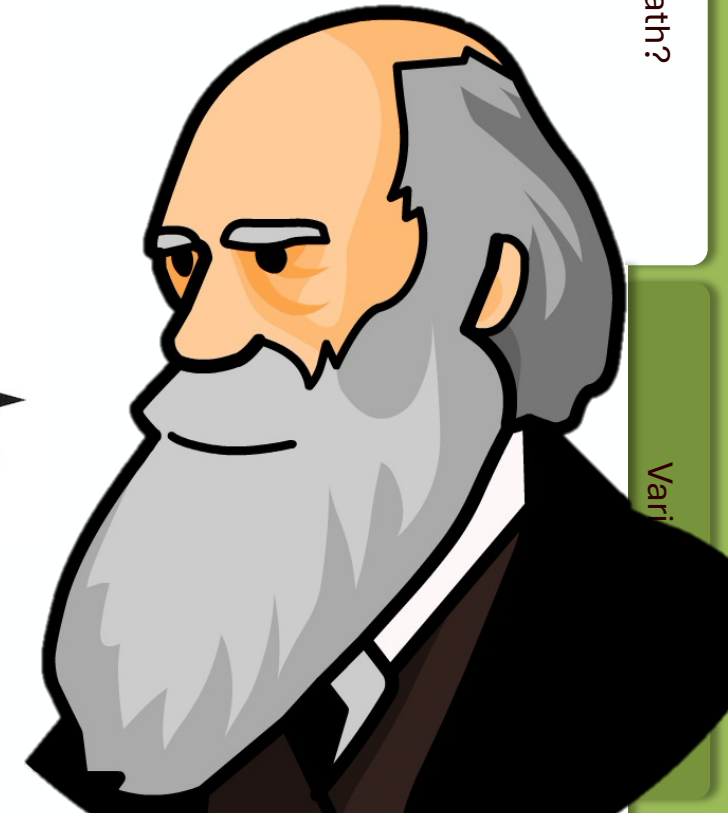
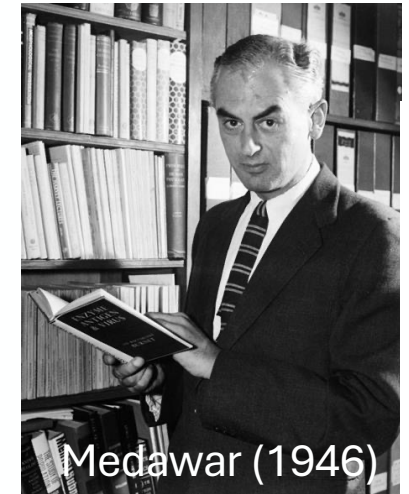
Since something will probably kill you anyways





How does this lead to senescence?

because strength of selection decreases, no mechanisms to counter mutations evolves





Why do mutations persist?
Tradeoffs, as per usual

Maintenance

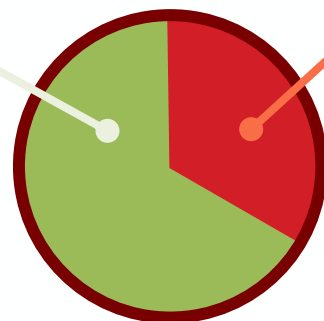
Reproduction





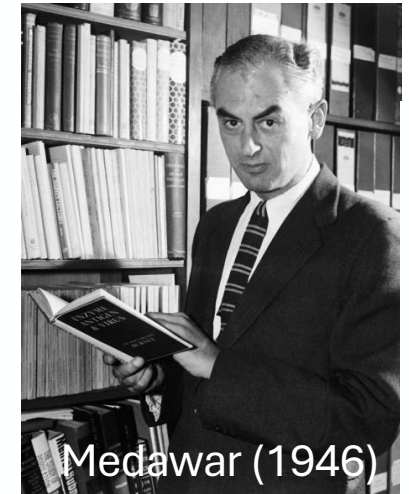
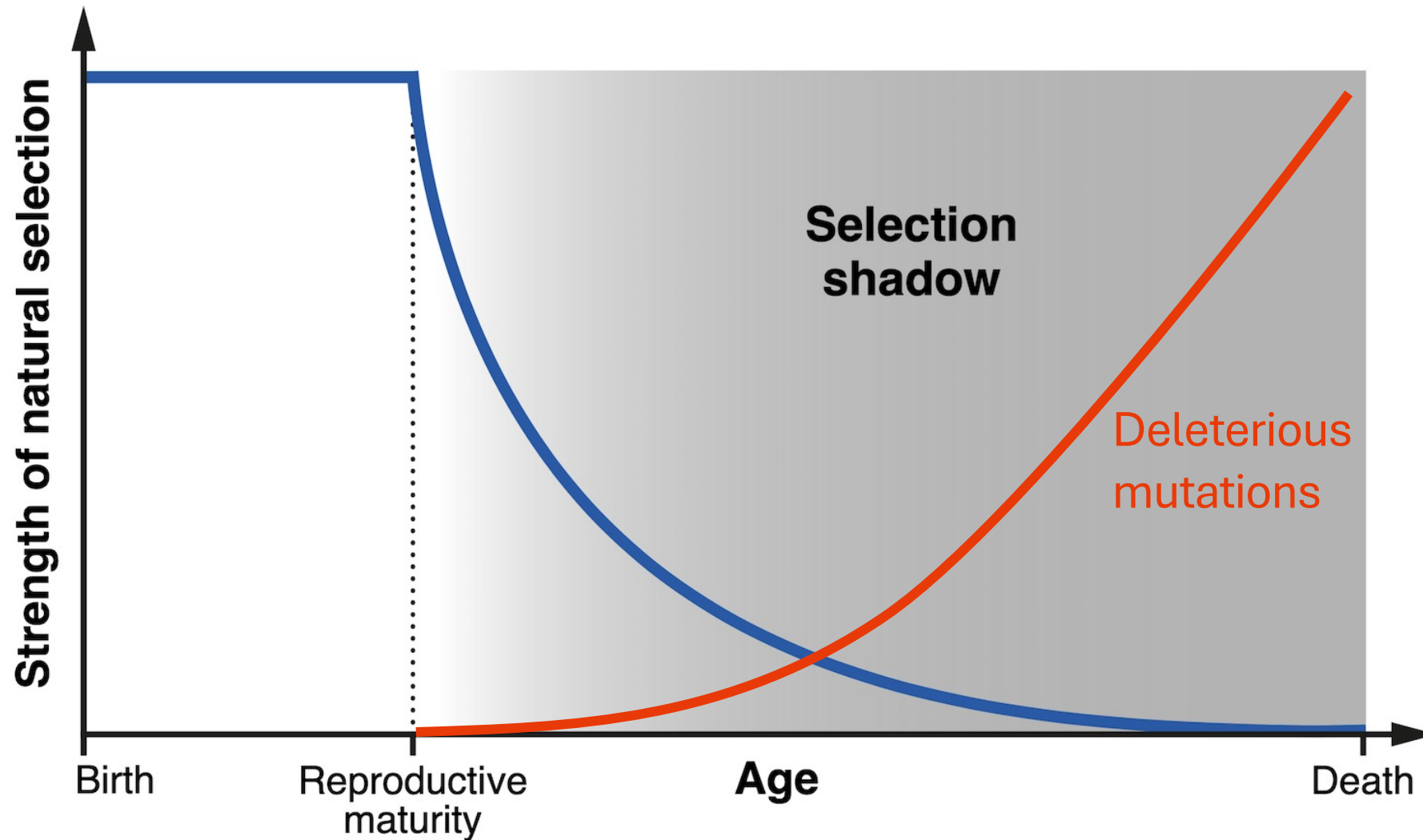
Why do mutations persist?
Tradeoffs, as per usual

Maintenance



*Energy not used
for cellular repair*

Can't put
all energy into repairing
mutations b/c need to
reproduce once able



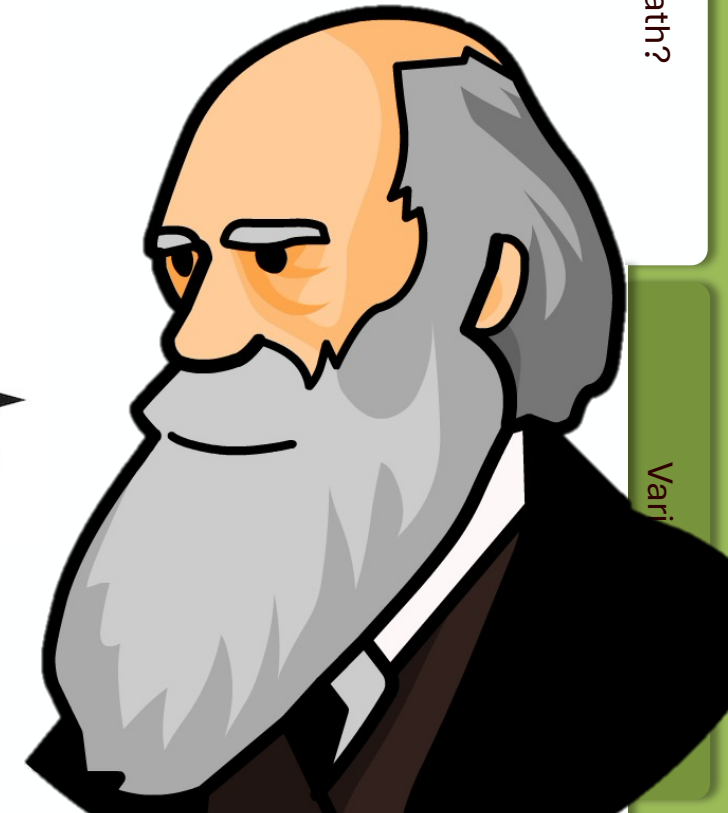
Intro

How death?

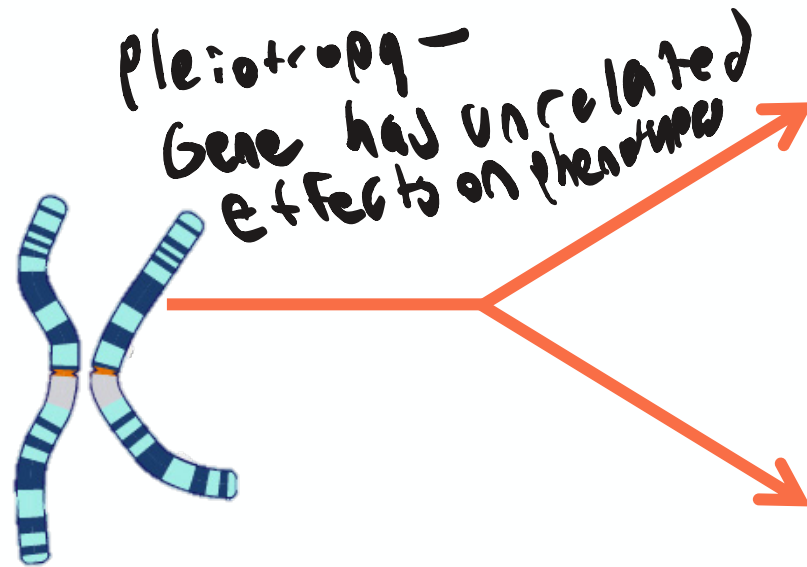
Vari

How does this lead to senescence?

because strength of selection decreases, no mechanisms to counter mutations evolves



Antagonistic pleiotropy

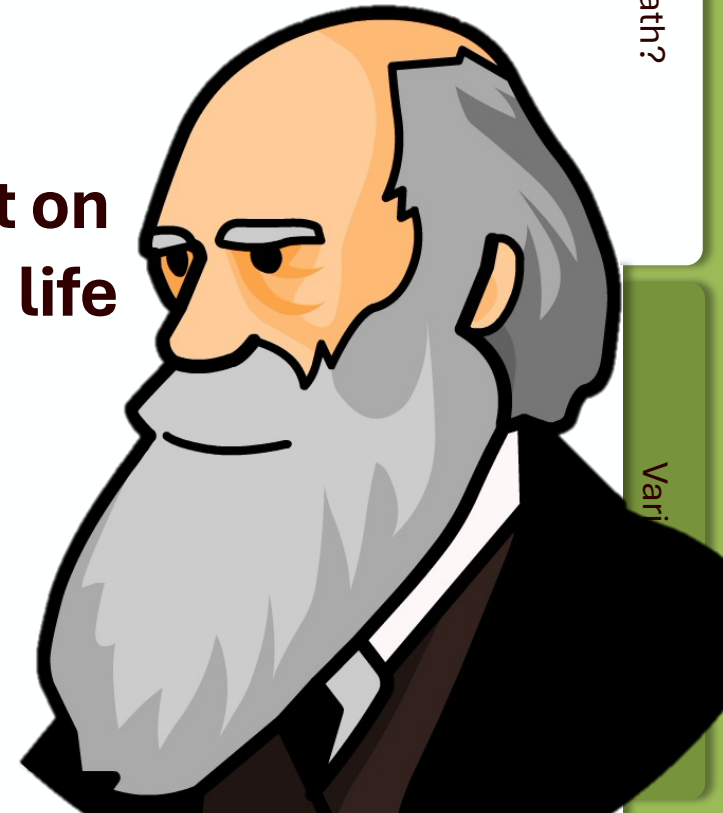


Trait 1

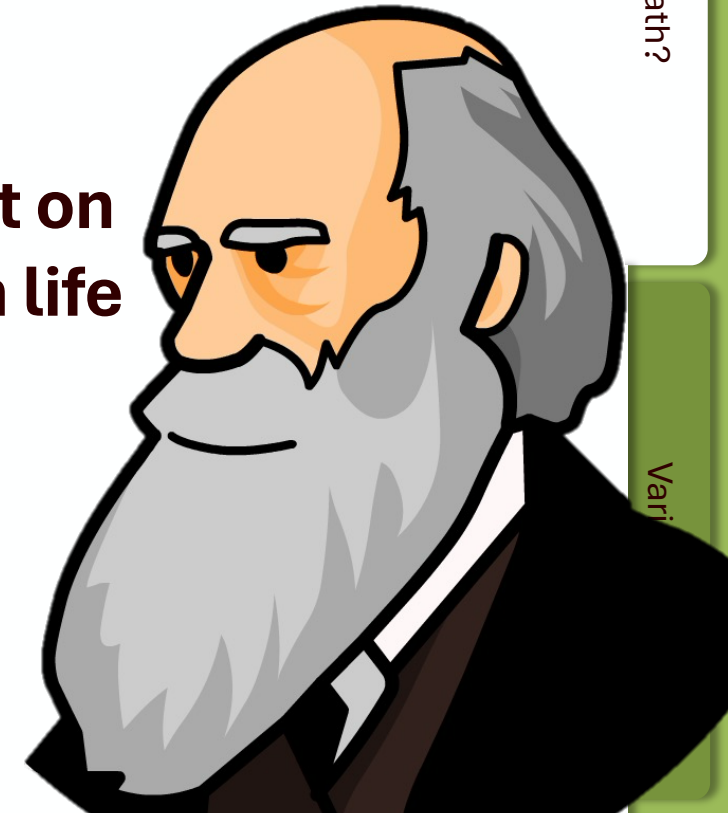
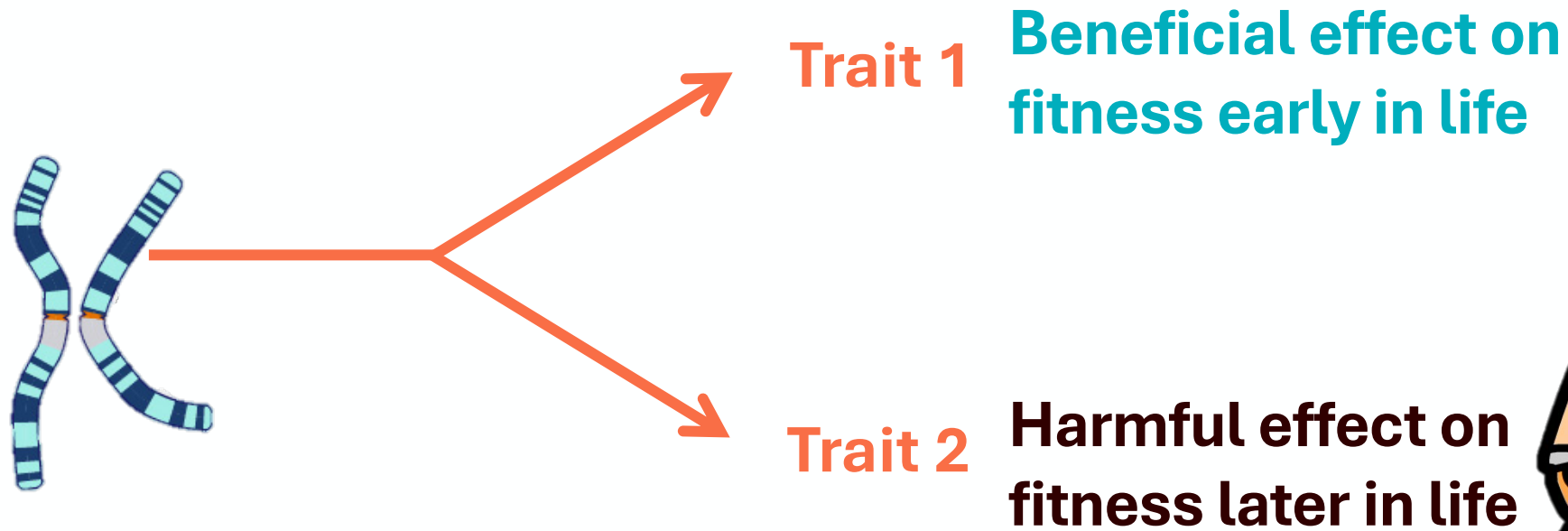
**Beneficial effect on
fitness early in life**

Trait 2

**Harmful effect on
fitness later in life**



Antagonistic pleiotropy

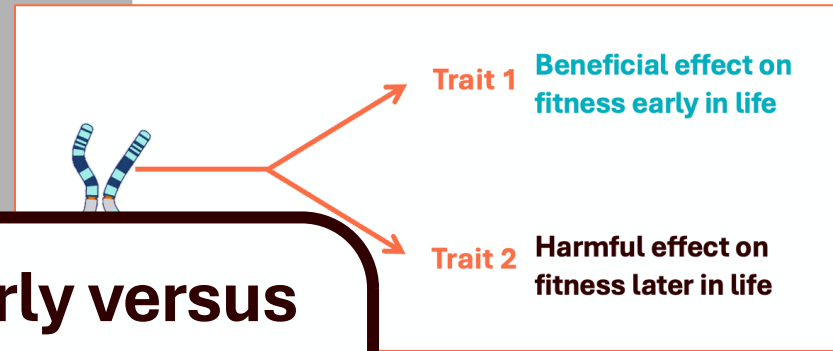




Strength
natural

Selection
shadow

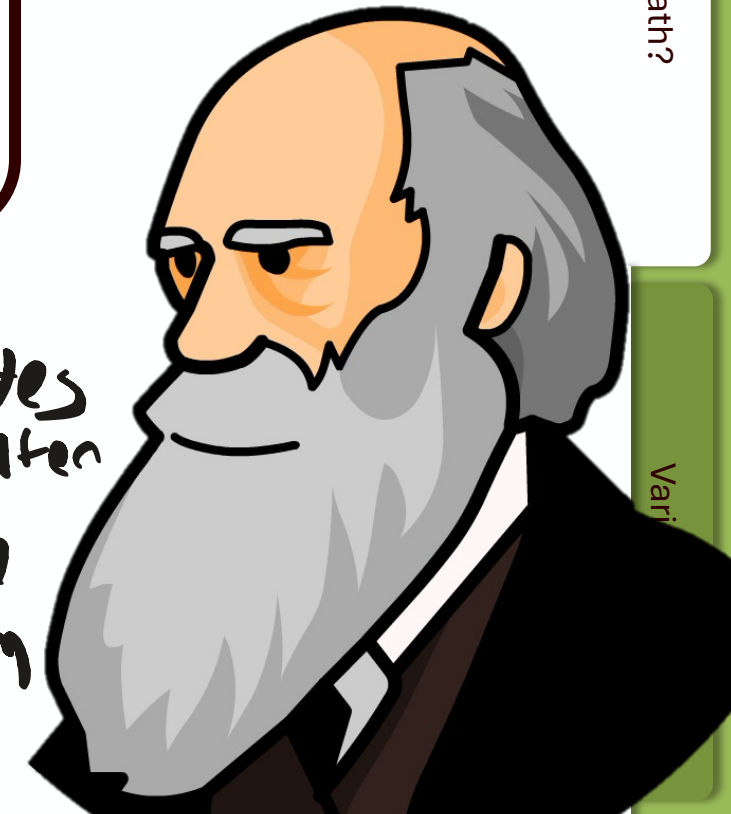
Tradeoff between early versus late reproduction is due to **tradeoff** between reproduction and maintenance



Birth Reproductive maturity Age Death

Something is wrong w/ wording, check slides later
because strength of selection decreases, no selection against genes that have negative effects later in life... and there will be positive selection if they have beneficial effects early in life

add them

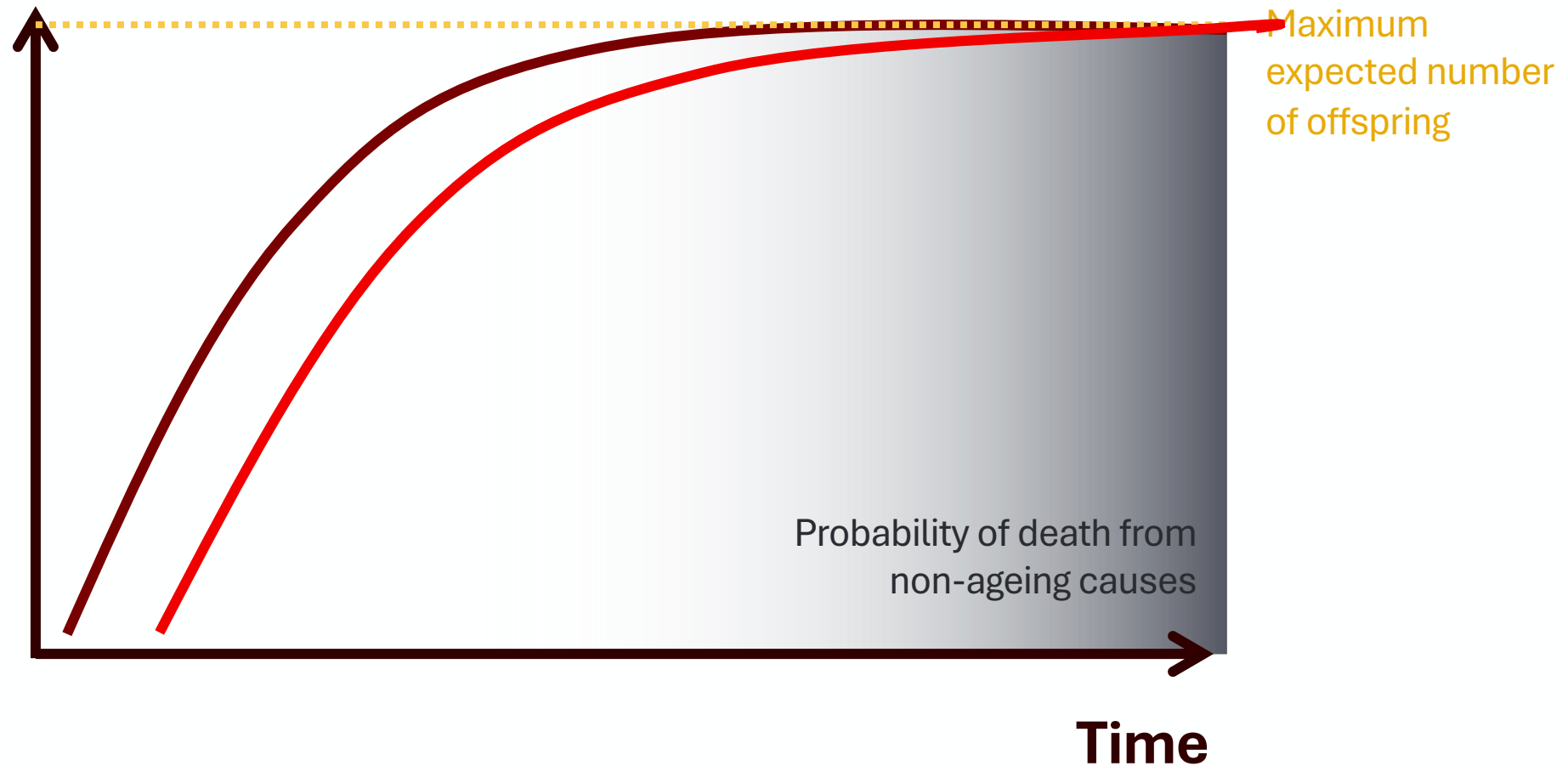


Intro

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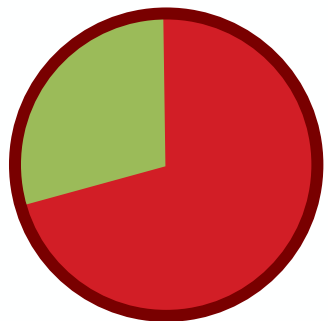
Vari

Early reproduction:



Processes that lead to senescence

Offspring

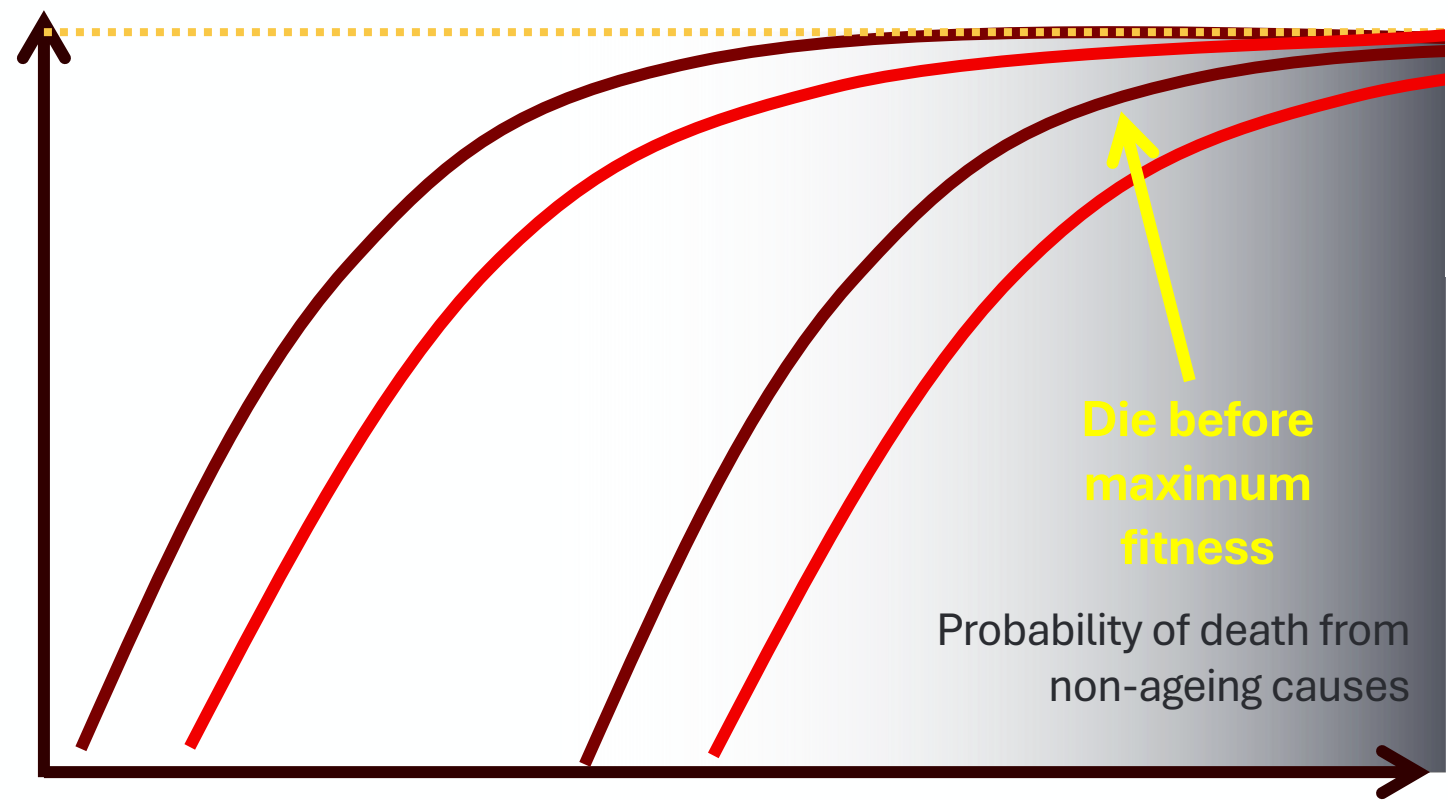
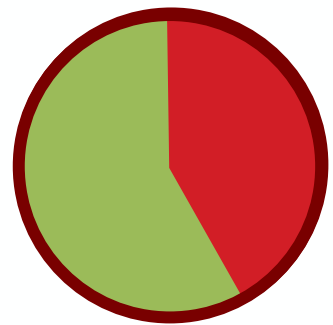


Later reproduction reduces effects of senescence and aging

To reduce processes that lead to senescence...

Processes that lead to senescence

Offspring



Die before maximum fitness

Probability of death from non-ageing causes

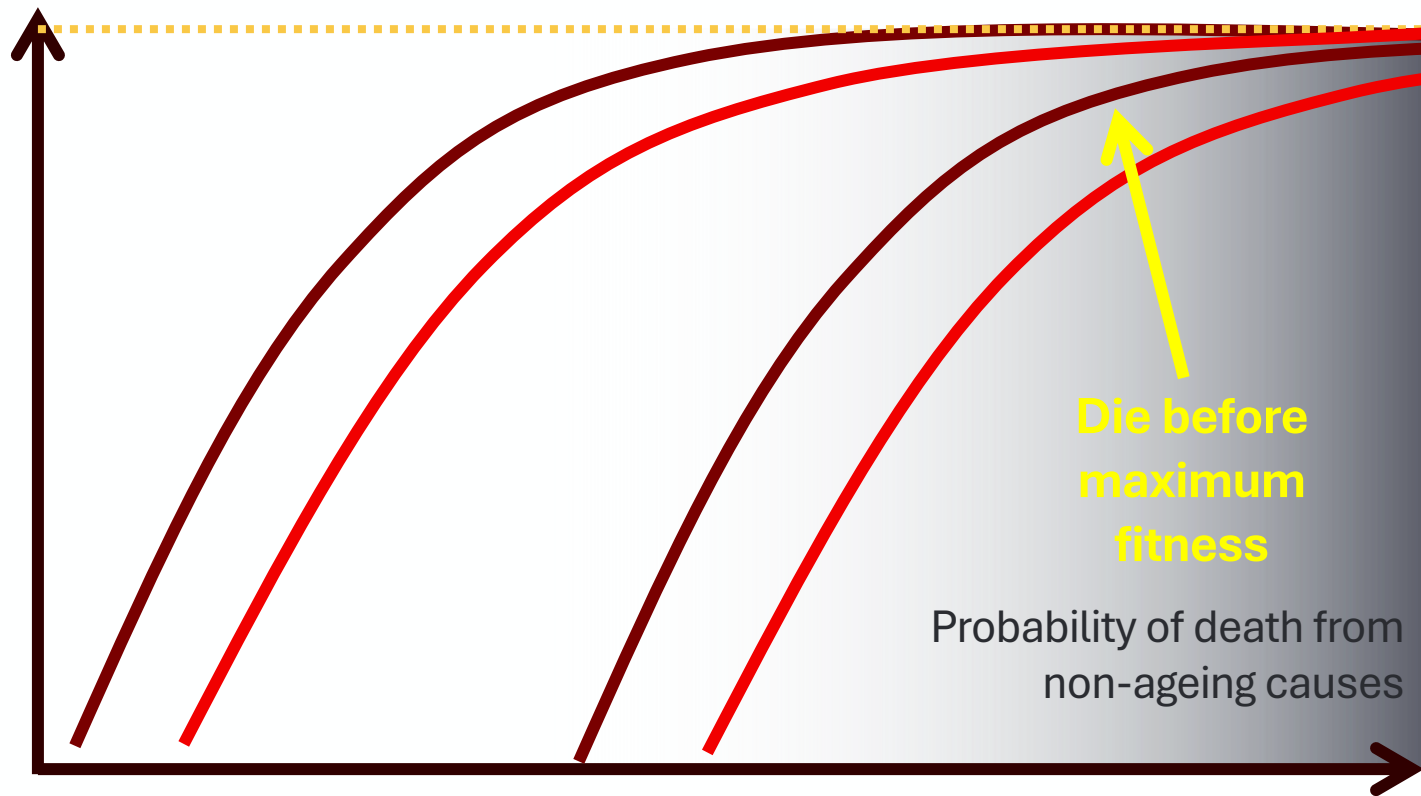
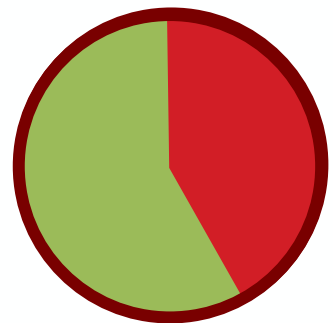
Time

Maximum expected number of offspring
but due to higher level of extrinsic mortality you might not have expected # of offspring

To reduce processes that lead to senescence...

Processes
that lead to
senescence

Offspring



Maximum
expected number
of offspring

Die before
maximum
fitness

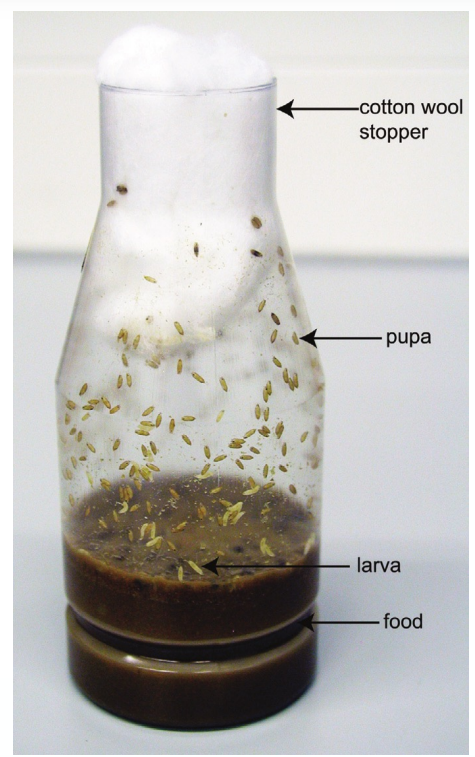
Probability of death from
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Time

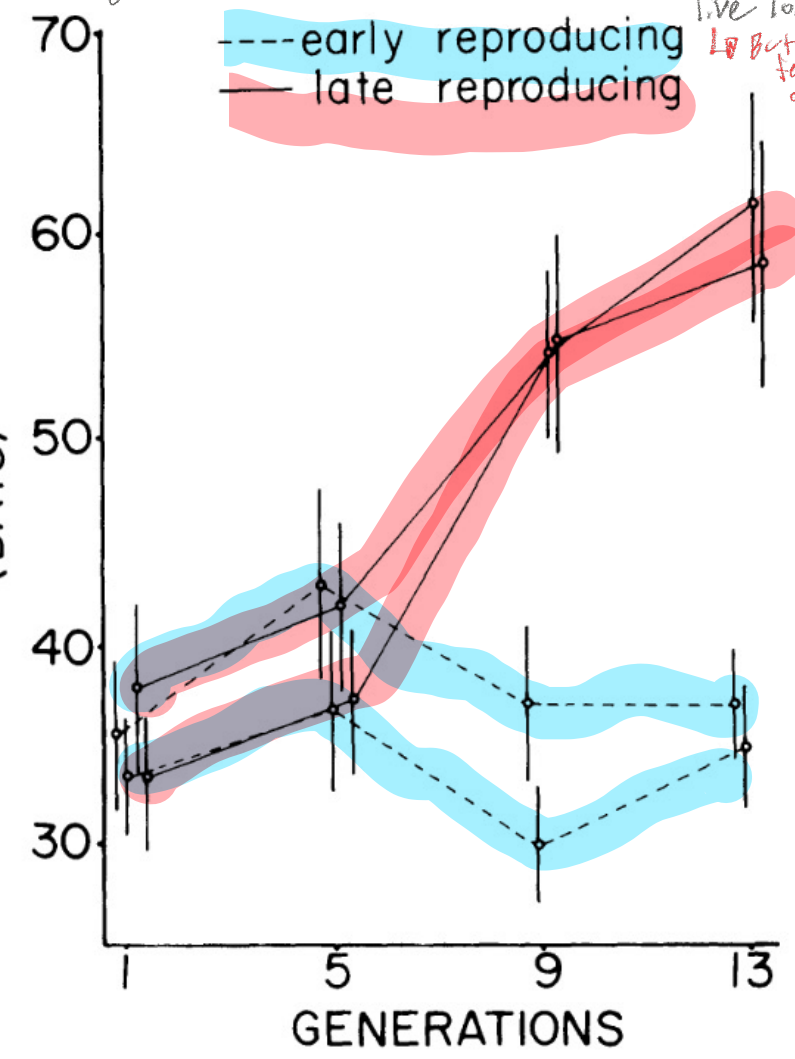
Evolution, 38(5), 1984, pp. 996-1003

SELECTION FOR DELAYED SENESCENCE IN DROSOPHILA MELANOGASTER

LEO S. LUCKINBILL, ROBERT ARKING, MICHAEL J. CLARE,
WILLIAM C. CIROCCO, AND STEVEN A. BUCK

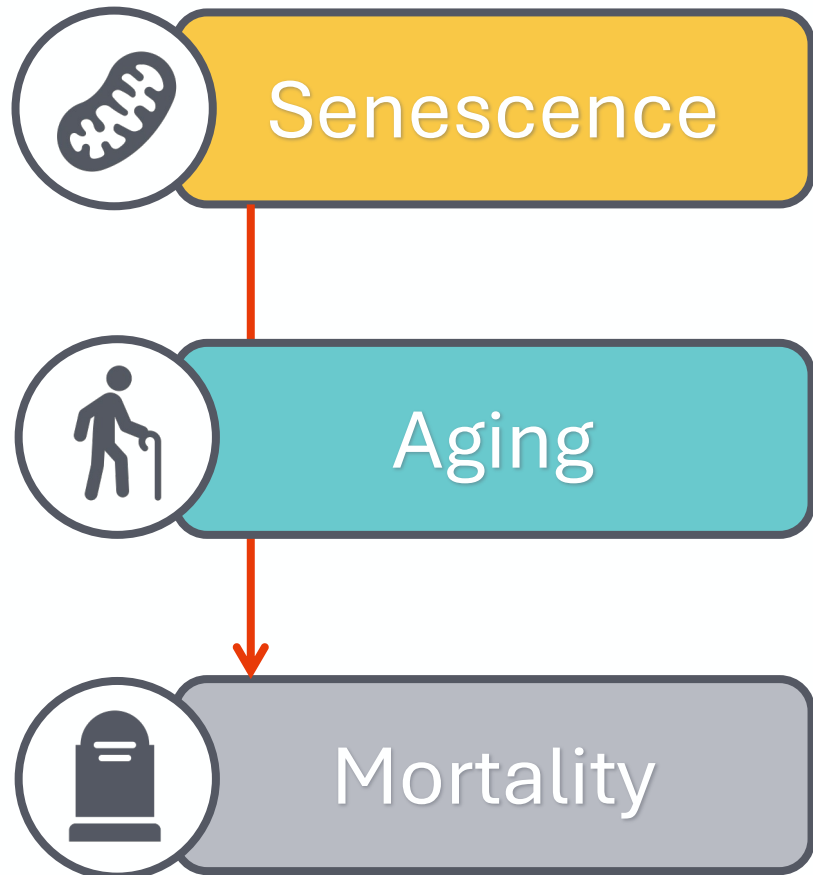


AVERAGE LONGEVITY (DAYS)



Early reproducing → longevity decreases
Late reproducing → more energy into cellular repair → live longer

But fewer offspring
↳ trade-off between maintenance + reproduction



Some of the processes that underlie senescence and ageing:

1. Telomere attrition: telomeres shortening
2. Genomic instability: Excessive DNA damage or insufficient DNA repair
3. Epigenetic alterations: Alterations in the methylation of DNA or acetylation and methylation of histones, as well as of other chromatin-associated proteins
4. Loss of proteostasis: Failure to refold or degrade unfolded proteins
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7. Stem cell exhaustion
8. Altered intercellular communication: changes at the level of intercellular communication, be it endocrine, neuroendocrine or neuronal

The biocultural determinants of variability in ageing and senescence

The biocultural determinants of variability in ageing and senescence

1. Age-specific mortality rate
2. Lifespan
3. Life expectancy
4. Longevity

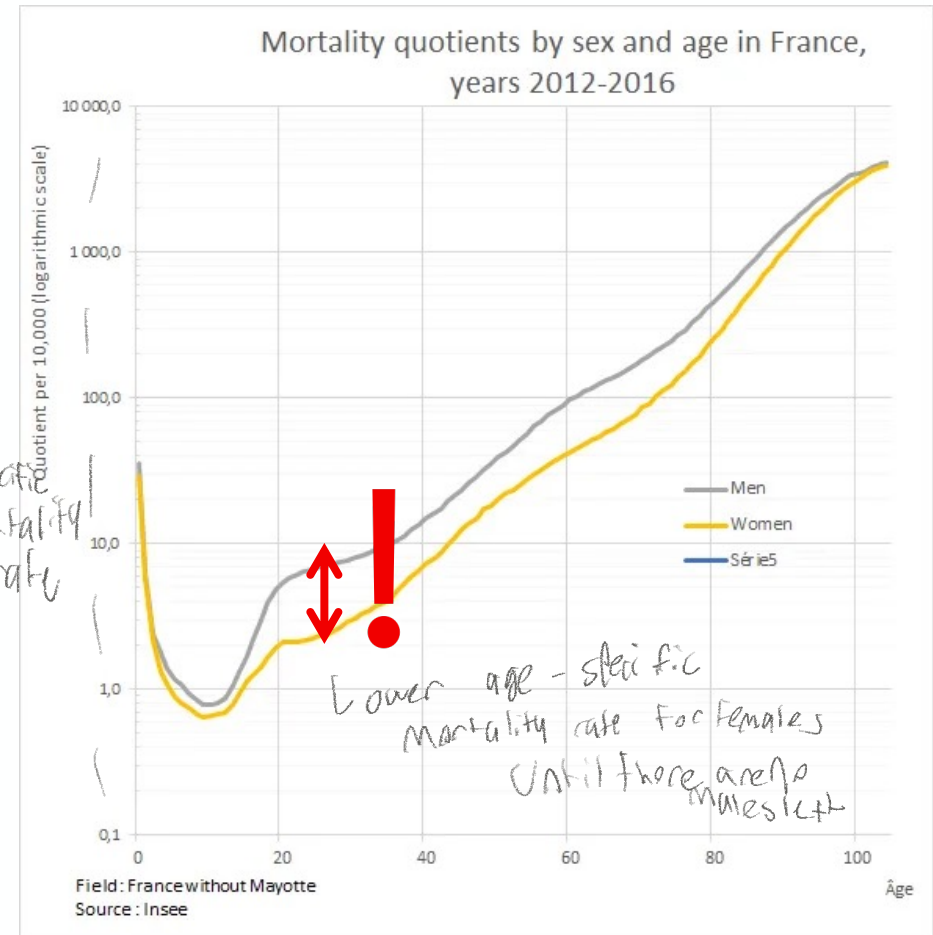
The biocultural determinants of variability in ageing and senescence

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Number of deaths in
age group

Number of people in
age group

Divided by

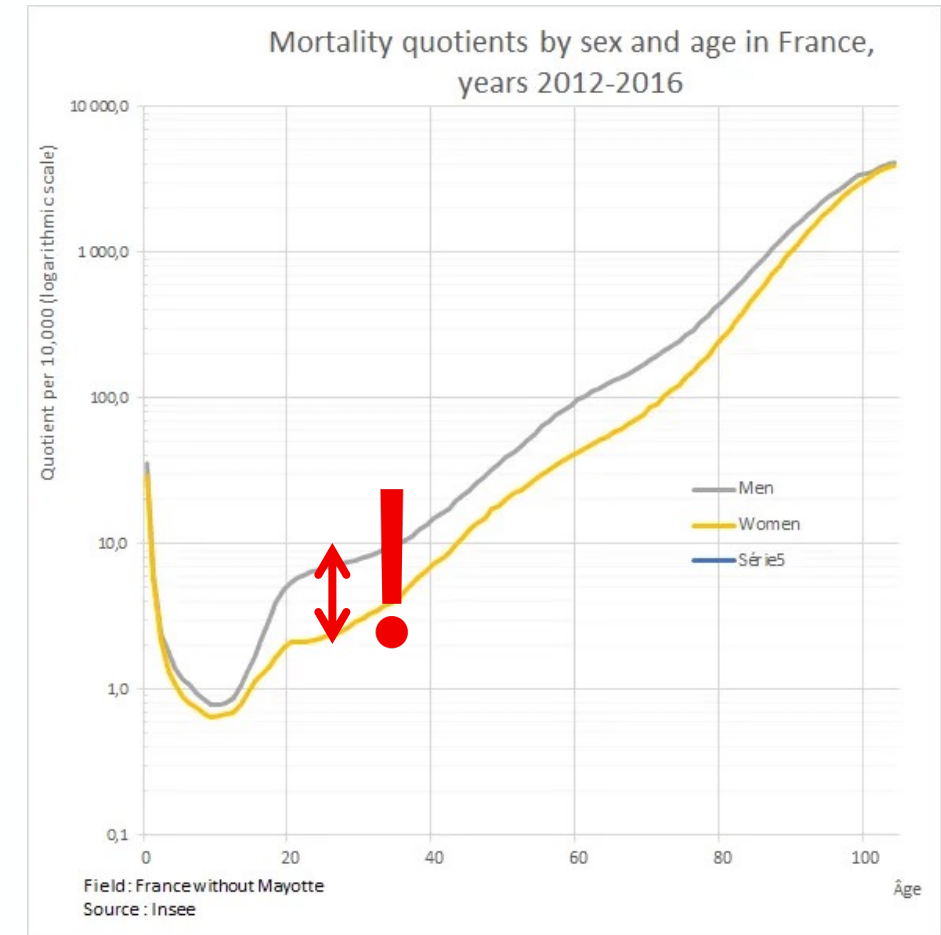


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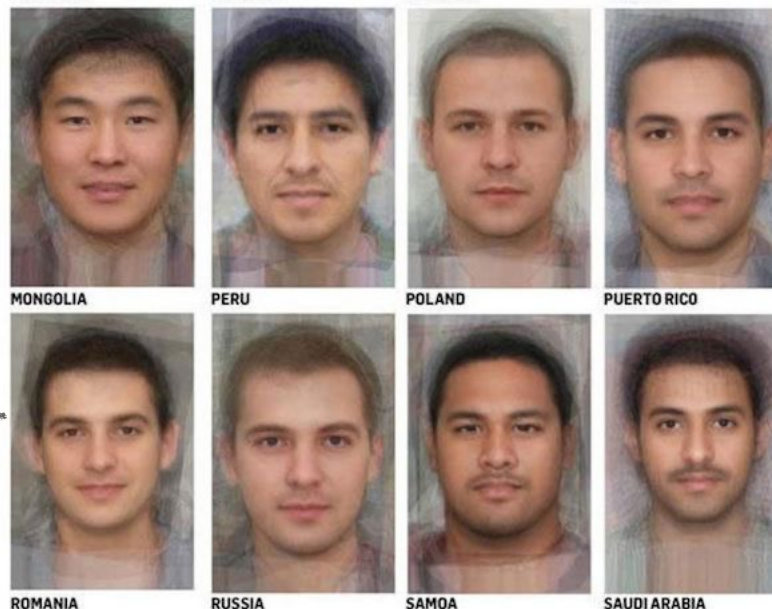


The biocultural determinants of variability in ageing and senescence

1. Age-specific mortality rate
2. **Lifespan**
3. Life expectancy
4. Longevity

The **typical** duration of existence of an individual member of a species

Just
an
average,
averages
are
abstraction



- 57% of all those ages 65 and older are female. By age 85, 67% are women.
- USA: females live 5 years longer than males
- Worldwide: females live 7 years longer



The biocultural determinants of variability in ageing and senescence

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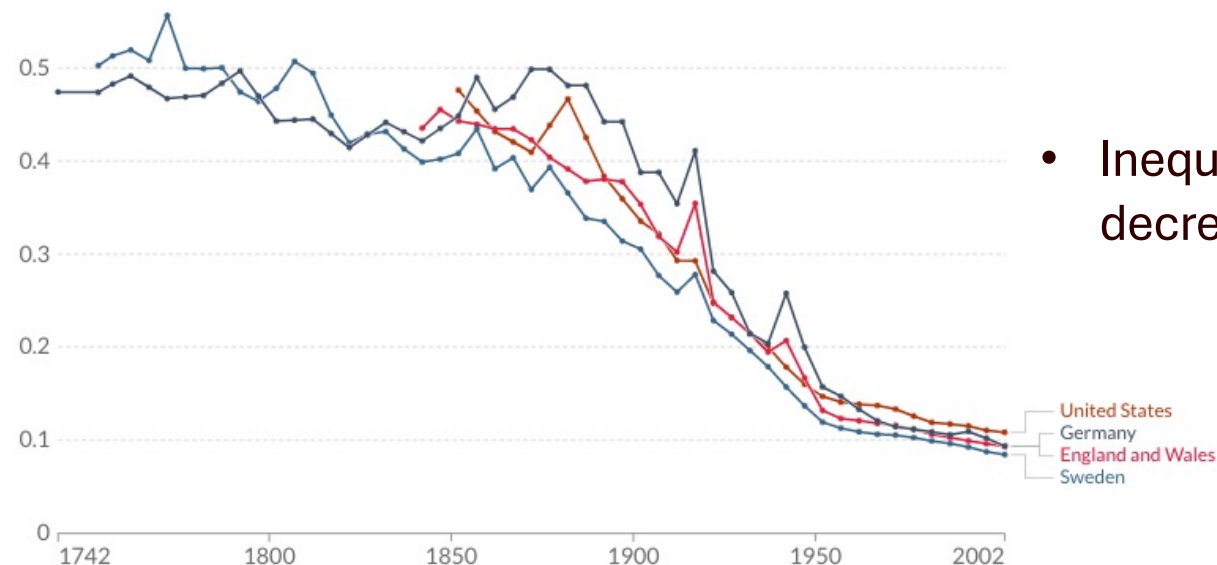
The **typical** duration of existence of an individual member of a species

Inequality of life years, 1742 to 2002

The inequality of years of life between people within the same country is measured with the Gini index here. A high Gini means large within-country inequalities of the number of years that people live.

Our World
in Data

+ Add country



- Inequality in lifespan has decreased over time

The biocultural determinants of variability in ageing and senescence

1. Age-specific mortality rate
2. Lifespan
3. **Life expectancy**
4. Longevity

Characteristic of populations and corresponds to the **expectation of life at any given age.**

Life expectancy at birth is the average life span resulting from all-cause mortality of a cohort (i.e., a group of individuals born in the same year)

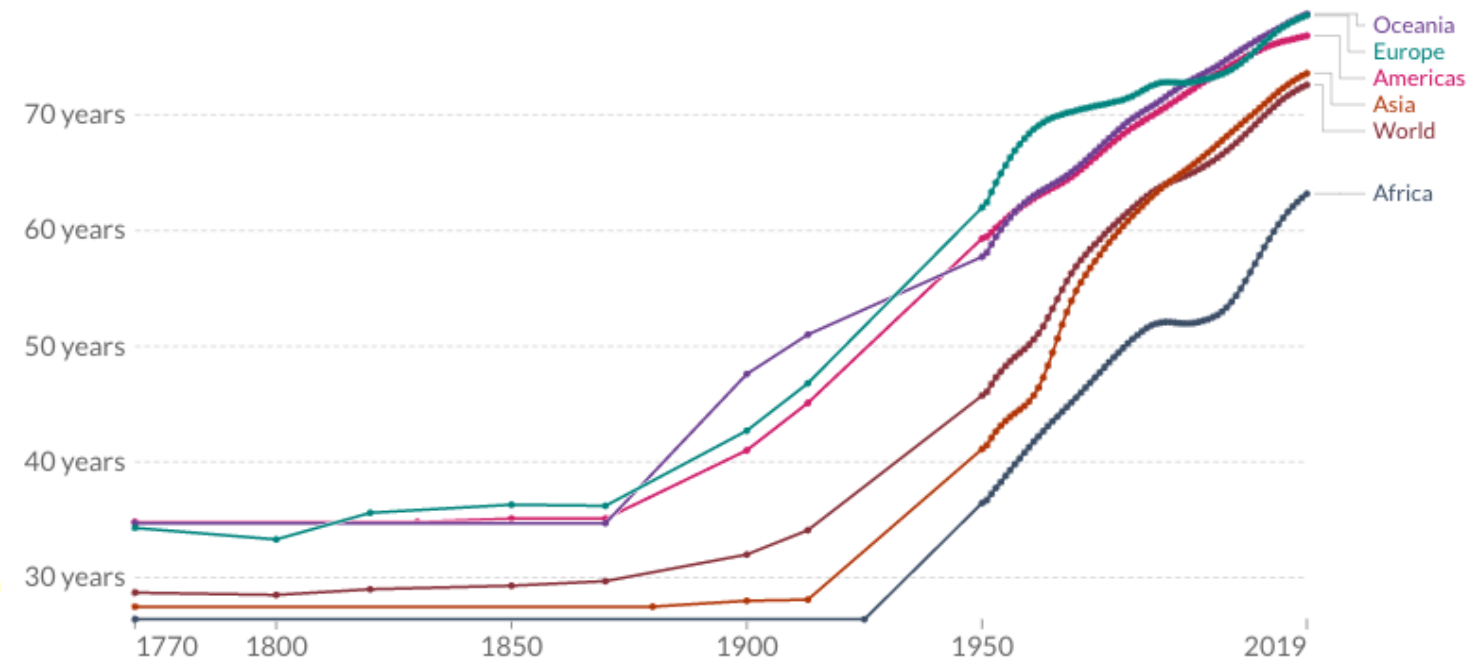
The biocultural determinants of variability in ageing and senescence

1. Age-specific mortality rate
2. Lifespan
- 3. Life expectancy**
4. Longevity

- Life expectancy at birth has increased over time

Life expectancy, 1770 to 2019

LINEAR LOG + Add country



Source: Riley (2005), Clio Infra (2015), and UN Population Division (2019)
Note: Shown is period life expectancy at birth, the average number of years a newborn would live if the pattern of mortality in the given year were to stay the same throughout its life.

OurWorldInData.org/life-expectancy • CC BY

The biocultural determinants of variability in ageing and senescence

1. Age-specific mortality rate
2. Lifespan
3. Life expectancy
4. **Longevity**

an individual phenomenon; the duration of an individual life

The biocultural determinants of variability in ageing and senescence

1. Age-specific mortality rate
2. Lifespan
3. Life expectancy
4. Longevity

Summary of patterns:

- Females can expect to live longer than males
- Life expectancy has increased over time
- Inequality in lifespan has decreased over time



Which of these patterns reflects changes or differences in the rate of senescence?

The biocultural determinants of variability in ageing and senescence

1.
2.
3.
4.



- Summary of patterns:
- Life expectancy gains in ~1900-1950: due to improvements in **infant and child survivorship**
 - Life expectancy gains in ~1950-2000: **adult mortality decline**, including late age mortality
 - Falling tobacco use (for men only) and cardiovascular disease mortality (for both men and women)
 - **Life expectancy has increased over time**
 - Inequality in lifespan has decreased over time
- Due to* (handwritten note with an arrow pointing to the adult mortality decline bullet)



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The biocultural determinants of variability in ageing and senescence

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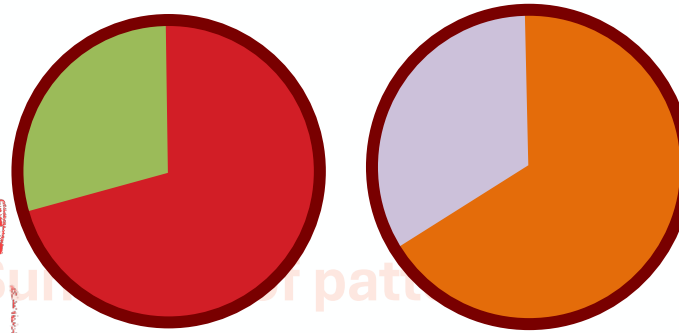
The biocultural determinants of variability in ageing and senescence

Sexual selection hypothesis

(Kruger and Nesse, 2004)

Males: selected for success in reproductive competition at the expense of health and longevity

- Increased risk from greater vulnerability of male structural, physiological, endocrinological, and immunological systems, especially lower resistance to infection, injury, stress and degenerative diseases
- Compared to females, males tend to have greater height and weight, more upper-body strength, and higher metabolic rates



- Females can expect to live longer

Less maintenance?

- Inequality in lifespan

Slow and steady wins the race



Which of these patterns reflects changes or differences in the rate of senescence?



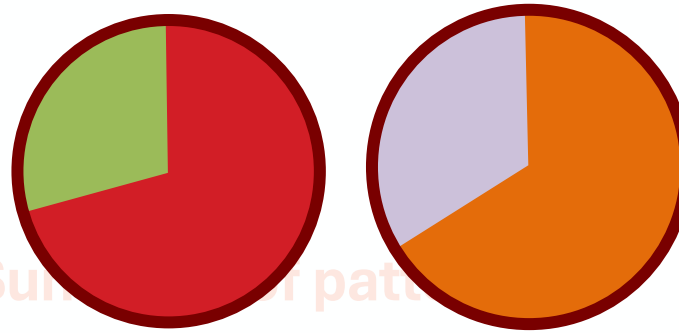
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- **Females can expect to live longer**

- Life expectancy has increased over time

- Inequality in lifespan

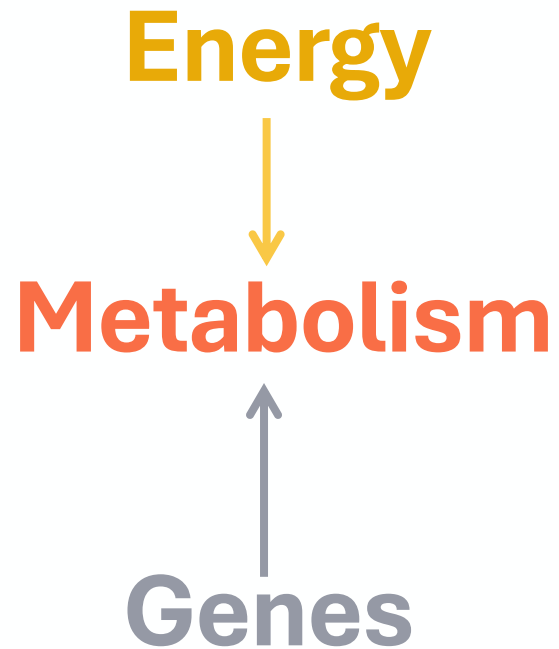
Slow and steady wins the race



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Living... is bad for you!

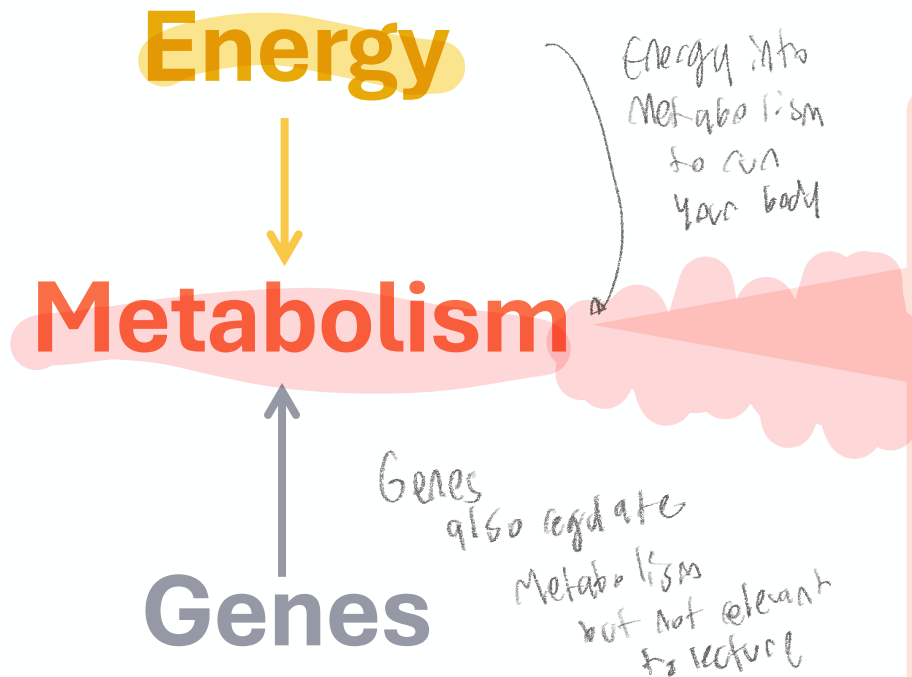


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8. Altered intercellular communication: changes at the level of intercellular communication, be it endocrine, neuroendocrine or neuronal

Prediction:

Slowing down metabolism, by reducing energy intake, should slow down senescence

Living... is bad for you!



Prediction:

Slowing down metabolism, by reducing energy intake, should slow down senescence

1. Telomere attrition: telomeres shortening
2. Genomic instability: Excessive DNA damage or insufficient DNA repair
3. Epigenetic alterations: Alterations in the methylation of DNA or acetylation and methylation of histones, as well as of other chromatin-associated proteins
4. Loss of proteostasis: Failure to refold or degrade unfolded proteins
5. Deregulated nutrient sensing
6. Mitochondrial dysfunction
7. Stem cell exhaustion
8. Altered intercellular communication: changes at the level of intercellular communication, be it endocrine, neuroendocrine or neuronal

Living... is bad for you!

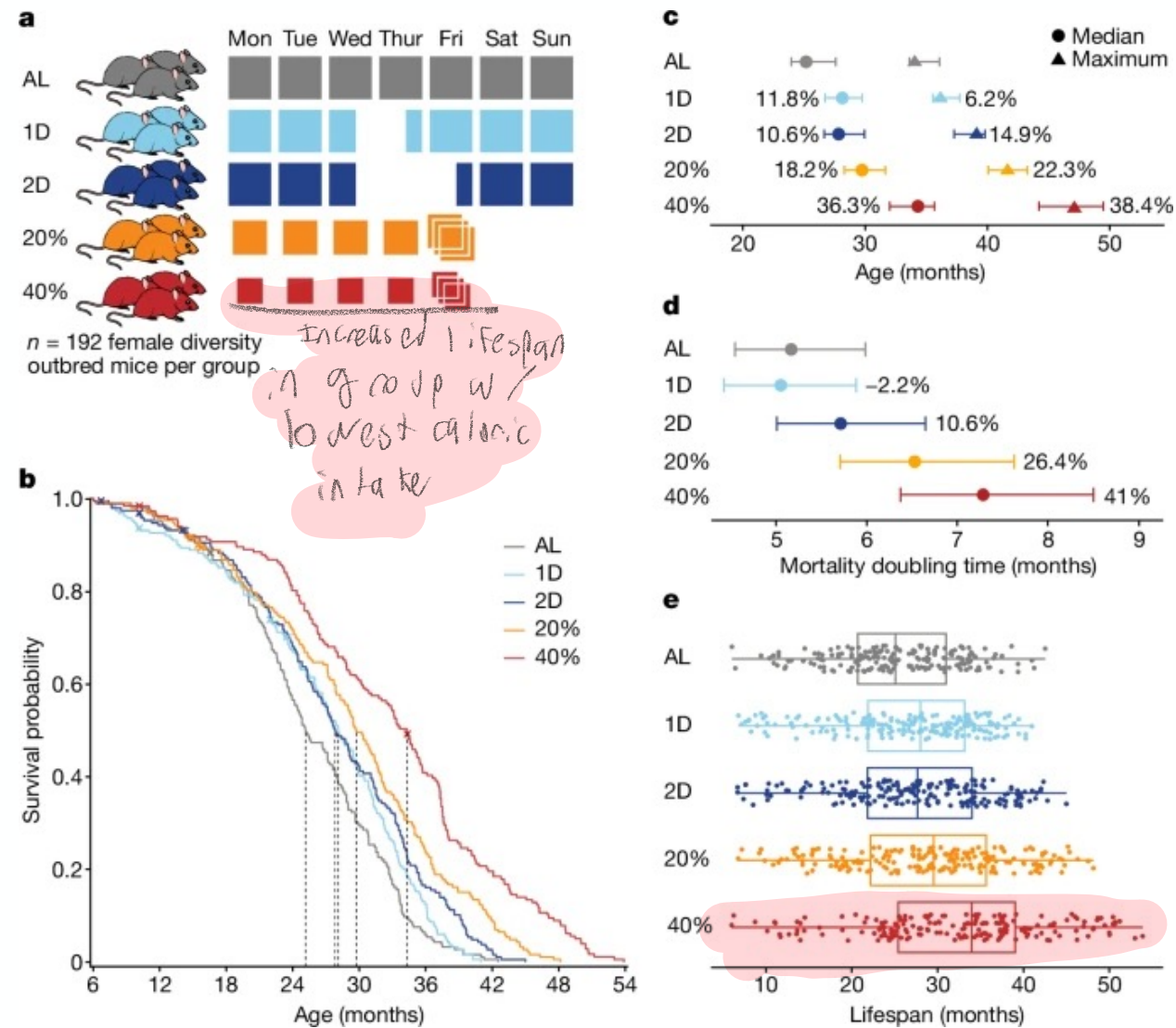
Intermittent fasting

Caloric restriction



Prediction:

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Living... is bad for you!

nature aging



Letter

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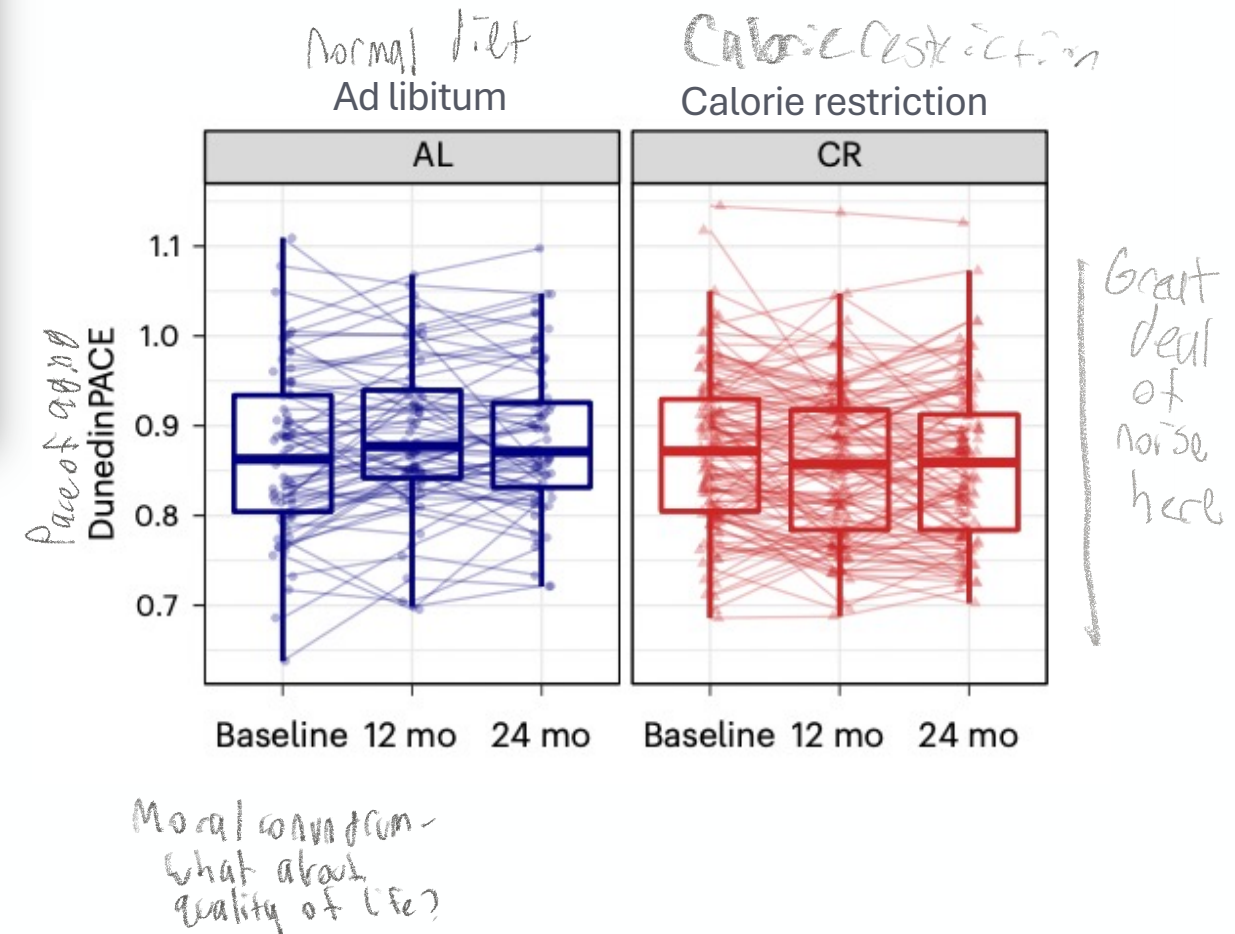
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Check for updates

R. Waziry^{1,16}, C. P. Ryan^{1,16}, D. L. Corcoran², K. M. Huffman³, M. S. Kobor⁴, M. Kothari¹, G. H. Graf^{1,5}, V. B. Kraus³, W. E. Kraus³, D. T. S. Lin⁴, C. F. Pieper⁶, M. E. Ramaker³, M. Bhapkar⁷, S. K. Das⁸, L. Ferrucci⁹, W. J. Hastings¹⁰, M. Kebbe¹¹, D. C. Parker¹², S. B. Racette^{13,14}, I. Shalev¹⁰, B. Schilling¹⁵ & D. W. Belsky^{1,5} ✉

Prediction:

Slowing down metabolism, by reducing energy intake, should slow down senescence





- The current understanding is the death is a (non-adaptive) by-product of senescence and ageing.
- The reason that natural selection has not selected for processes that reverse senescence and ageing, or prevent them in the first place, is the selection shadow.
- In addition, early reproduction is always favoured in the tradeoff between early and late reproduction. *Increases fitness but earlier senescence*
- The factors that lead to senescence are by-products or inevitable mistakes that accumulate during metabolism. Slowing metabolism has been shown to reduce the rate of senescence.